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Technical paper

Machining parameter optimization for a batch milling system using multi-task deep reinforcement learning

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ABSTRACT

The integrated multi-objective optimization of machining parameters for improved machining quality and efficiency is important in batch milling systems. Due to the change of the batch milling system state, the continuous use of the same machining parameters may lead to degradation in quality and efficiency for workpieces in batches. Machining parameter optimization is usually determined by manual experience or trial-and-error methods, making it difficult to achieve a synergistic consideration of both quality and efficiency. To address this issue, a novel multi-task deep reinforcement learning method for machining parameter optimization in a batch machining system is proposed. Firstly, a reliable parallel joint estimation model of multiple machining quality and efficiency indicators is established using a multi-task time series estimation method, which can learn the correlation of these indicators to improve estimation accuracy. Then, the parameter optimization problem is formalized as a Markov decision process supported by a reinforcement learning virtual environment and an agent. The reinforcement learning virtual environment with the joint estimation model is constructed to improve the accuracy of optimized machining parameters for the collaborative optimization of quality and efficiency indicators. Within the virtual environment, time series sequential state, sequential action, multi-objective reward function, and constraint conditions adapted to the joint estimation model are defined to repeatedly evaluate different machining parameters. The agent with a multi-head attention and a dynamic weight adjustment mechanism is designed to improve the stability of the optimization process. Finally, experiments on a real machining dataset of thin-walled parts show that compared with the traditional deep reinforcement learning algorithm, the optimization effect of the proposed framework is improved by 9 %–12 %, and the standard deviation is decreased by 9 % –18 %.

1. Introduction

As a common form of production, milling is often used for batch machining of difficult-to-machine workpieces due to its advantages of high quality and efficiency. The structure and quality requirements for batch machining of multiple workpieces are consistent. In order to ensure the quality and efficiency requirements of each workpiece, production adjustments are allowed during the batch machining process. If the same machining parameters are continuously used, the batch machining quality of the same or similar workpieces may deteriorate, leading to the reduction in efficiency (such as machining time) and affecting the consistency of machining quality (such as dimensional errors). Therefore, the machining parameters between workpieces in batch milling are variable. Especially for the milling machining of multi-machining feature workpieces, the contradiction between quality and efficiency is more prominent [1]. Consequently, how to simultaneously improve their machining quality and efficiency in multi-objective optimization is a challenge [2]. With the development of big data and sensing technology, data-driven methods have attracted widespread attention due to their ability to solve analytical problems in a personalized, customized, and highly targeted manner based on more raw production data without requiring complex physical models [2,3]. They have been applied to the multi-objective optimization process of milling quality and efficiency, such as machining parameter optimization considering surface roughness and material removal rate.

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Nomenclature

DRL	Deep Reinforcement Learning.
MTE-BDRL	Multi-task time series, estimation and Balanced, Deep Reinforcement Learning.
DWA	Dynamic Weight Adjustment.
TD3	Twin Delayed Deep, Deterministic policy, gradient algorithm.
SVR	Support Vector Regression.
DNN	Deep Neural Networks.
CNN	Convolutional Neural, Networks.
MTL	Multi-Task Learning.
DQN	Deep Q-Network.
LSTM	Long Short-Term Memory.
DDQN	Double Deep Q-Network.
A3C	Asynchronous Advantage, Actor-Critic.
GNN	Graph Neural Networks.
PCA	Principal Component Analysis.
MMoE	Multi-gate Mixture of Experts.
RCNN	Residuals Convolutional, Neural Network.
BiLSTM	Bi-directional Long, Short-Term Memory.
MDP	Markov Decision Process.
BMMoE-TE	Bi-directional Multi-gate, Mixture of Experts time-series estimation.
R^2	R-square.
MAE	Mean Absolute Error.
RMSE	Root Mean Square Error.
MATD3	Multi-head Attention Twin, Delayed Deep Deterministic, policy gradient.
OU	Ornstein-Uhlenbeck.
GMME-TE	Gate Multi-gate Mixture of, Experts time series estimation.
MMoE-TE	Multi-gate Mixture of Experts, time series estimation.
RNN	Recurrent Neural Network.
GRU	Gate Recurrent Unit.
DDPG	Deep Deterministic Policy, Gradient.
MADDPG	Multi-head Deep Deterministic, Policy Gradient.
RTD3	Residuals Twin Delayed Deep, Deterministic policy gradient.
Std	standard deviation.

Although the existing methods in literature have made some progress in machining parameter optimization, existing methods usually focus on estimating multiple optimization objectives separately and neglect the correlation between coexisting objectives, resulting in low estimation accuracy. As shown in Fig. 1(a), the traditional optimization methods tend to build individual estimation model for each aim (i.e., objective) without considering the correlation between these aims. However, this is inconsistent with a real machining process, especially for thin-walled structures, as pursuing efficiency and economic benefits in complex and difficult-to-cut machining requires simultaneous attention to quality and efficiency. The optimized machining parameters supported by traditional optimization methods may produce poor application results in practice. The inadequate correlation and accuracy of multi-task estimation in iterative optimization environments results in suboptimal machining parameter optimization. Therefore, machining parameter optimization under multi-objective coupling remains a challenge.

Deep reinforcement learning (DRL) has been demonstrated as a feasible way to achieve multi-objective optimization for machining parameters with some success [4]. However, these existing reinforcement learning methods often ignore the correlation between optimization objectives, such as the mutual influence between dimensional quality and material removal efficiency of workpieces. The optimization objectives are established separately in the estimation model which is used to construct and update the virtual environment, which leads to the lack of environmental simulation effect.

Taking complex thin-walled parts as an example, this paper innovatively proposes a batch milling system machining parameter optimization method using multi-task deep reinforcement learning. The purpose of optimizing machining parameters is to improve multiple optimization objectives in parallel, that is, the machining quality (i.e., the dimensional quality indicators of machining features) and the machining efficiency (i.e., the machining time of each workpiece) of workpieces. Therefore, in this optimization method, a joint estimation model based on multi-task time series learning is constructed to form the virtual environment of the proposed multi-task deep reinforcement learning. The idea of the learning process of the proposed optimization method is briefly shown in Fig. 1(b). For the case of multiple optimization objectives in a complex production process, multi-task learning and deep reinforcement learning are fused to solve the problem of machining parameter optimization in time series batch production environments.

The main contributions of this paper are as follows:

1) Compared to traditional machining parameter optimization methods, a novel multi-task time series estimation and balanced deep reinforcement learning (MTE-BDRL) framework for machining parameter optimization is proposed in this paper. The framework efficiently addresses the issue of poor correlation among multiple aims through multi-task parallel estimation, enhancing the accuracy of multi-task estimation. By introducing the multi-task time series estimation model into the reinforcement learning environment, the accuracy of optimized machining parameters under the collaborative improvement of quality and efficiency multivariate machining indicators can be achieved.

2) Compared to traditional reinforcement learning methods, the virtual environment is mainly constructed based on multi-task parallel time series estimation and physical factors. This brings the virtual environment closer to the actual production process with multiple constraints under complex physical effects, ensuring better simulation performance. In addition, the multi-head attention mechanism and dynamic weight adjustment (DWA) are introduced in the twin delayed deep deterministic policy gradient algorithm (TD3), which improves the stability of the parameter optimization process.

The rest of the paper is as follows. Section 2 presents recent research advances in data-driven estimation methods and practical optimization problems in engineering. Section 3 introduces the specific implementation principle of the proposed model structure and the implementation method. In Section 4, the superiority of the proposed method is verified by multiple sets of comparative experiments. Finally, Section 5 summarizes the work of this paper and gives the future direction.

2. Related work

2.1. Related work of estimation model

The solution to machining optimization problems is generally achieved through estimation models of machining indicators (such as quality or efficiency). With the rapid advancement of machine learning and deep learning theories, numerous advanced techniques for estimating machining indicators have been proposed [5]. The initial estimation model is generally for the single task. Lee et al. [6] proposed a cyber-physical production system that combines support vector regression (SVR), decision tree, and random forest to enhance the prediction of processing quality in metal production. Wang et al. [7] combined residuals and Deep Neural Networks (DNN) to predict the quality and efficiency of CNC machining workpieces respectively. Li et al. [8] used

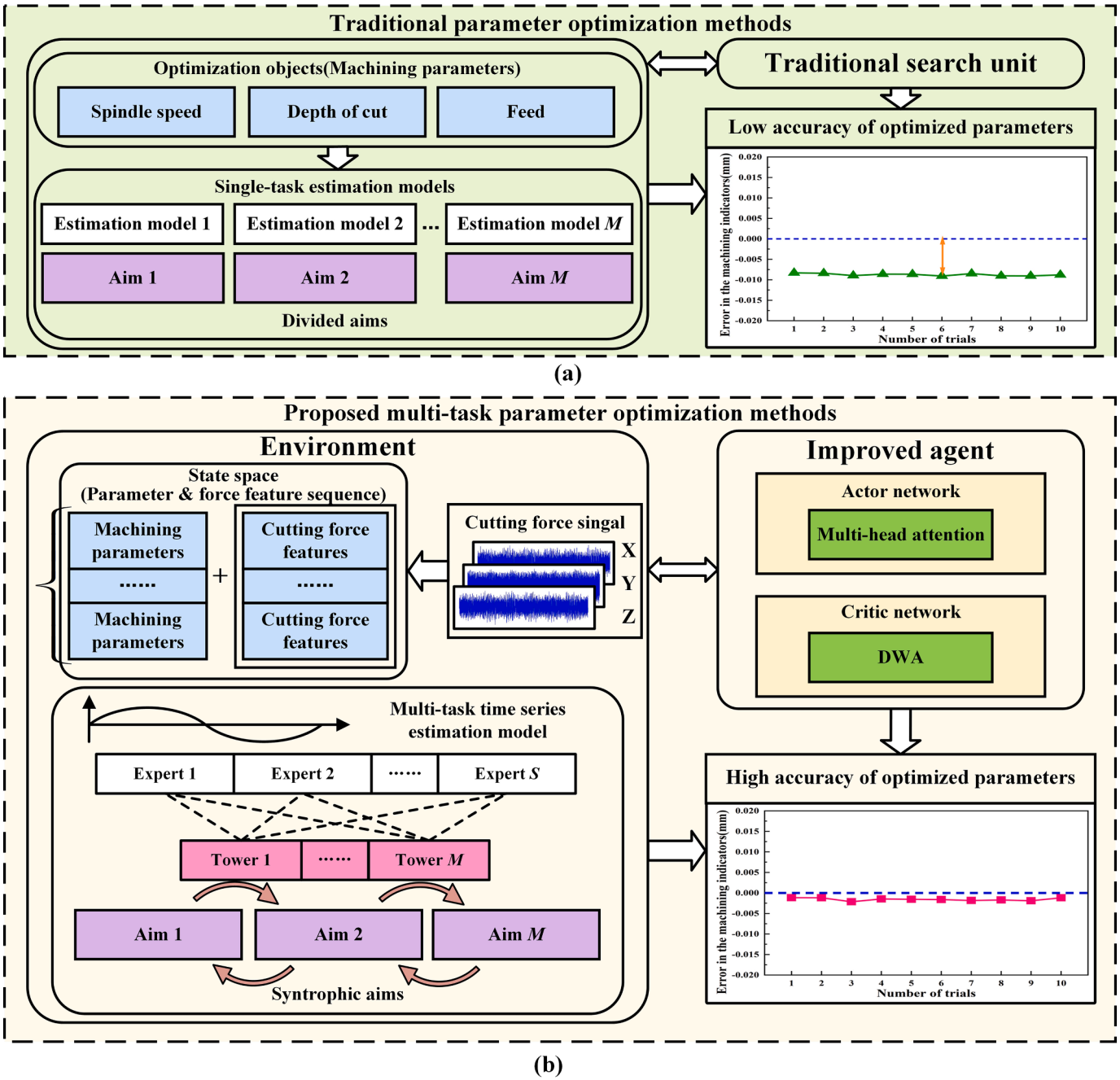

Fig. 1. Existing problems of parameter optimization and the main research contents of this paper. (a) Traditional milling machining parameter optimization methods; (b) Proposed multi-task milling machining parameter optimization methods.

Convolutional Neural Networks (CNN) to study the relationship between multi-dimensional process signals and quality under source operation conditions.

With more and more problems in industry that require multi-task parallel estimation, the research on intelligent manufacturing estimation models has gradually developed from single-task estimation to multi-task learning (MTL). The multi-task estimation model plays an important role in improving the accuracy of multivariate output because it can take into account the synergy between the estimation objectives. Sun et al. [9] proposed a random forest-based method to establish the relationship between critical variables and multiple quality indicators in the multi-stage production process. Chen et al. [10] developed a multi-task learning method to enable data-efficient spatiotemporal modeling using tool surface procession data in ultrasonic metal welding. Liu et al. [11] used multi-task learning with low-rank tensor constraints to estimate the deterioration forms of CNC tools. Wang et al. [12] performed assembly and testing quality prediction for multistage manufacturing systems using multi-task joint deep learning. Wang et al. [13] proposed a multi-task learning method based on the deep belief network, which can be used to predict tool wear and workpiece surface quality.

However, on the whole, these studies generally have two problems:

1) At present, the multi-task estimation model in the industrial scenario basically only takes the quality indicators as the estimation targets, but neglects the parallel prediction of quality and efficiency. However, for the current industrial demand, the machining quality and efficiency of products should be considered as optimization factors at the same time, and the correlation between the two should be concerned.

2) The joint estimation of multi-machining quality is not combined with multi-objective optimization, which may make it difficult for the estimation in multi-objective optimization to conform to production reality. Meanwhile, this could result in difficulties in reducing errors with optimization objectives during iterative optimization and render the optimization process unreliable.

2.2. Related work of optimization method

In recent years, there have been many studies in the area of machining parameters optimization for CNC machining [14]. Many of these studies have focused on methods based on mechanisms [15], machine learning approaches [16] or deep learning [7]. Both mechanistic research and simulation-based optimization techniques cannot adequately address the various factors affecting the optimization goal in complex machining problems [17]. Although machine learning algorithms introduce data-driven methods to handle multi-dimensional information, they have the limitation of a relatively random exploration process. It leads to the dilemma of falling into local optima in the process of solving optimization problems [18]. In addition, traditional machine learning methods still have shortcomings in computational efficiency and comprehensive information fusion when solving multi-objective collaborative improvement and machining parameter optimization problems in highly complex machining environments [19].

Reinforcement learning can handle complex environments and tasks by automatically interacting with the environment. And it can gradually improve the optimization policy through trial-and-error and feedback mechanisms. After the incorporation of deep neural networks, deep reinforcement learning has more advantages in dealing with nonlinear and high-level complex problems. As the original DRL algorithm, Deep Q-Network (DQN) successfully combines deep learning with reinforcement learning for the first time, realizing the ability to learn complex tasks in high-dimensional input space. Moreover, it exemplifies the transformative potential of DRL. Based on the DQN algorithm, many scholars have published valuable research on the optimization of machining parameters. Jiang et al. [20] proposed a LSTM-based DQN control policy for CNC machining. The policy utilizes a Long Short-Term Memory (LSTM) network to estimate contour error and enable enhanced machining compensation during the machining. Liu et al. [21] utilized meta-DQN to optimize the finishing process and control workpiece deformation. By improving Double deep Q-Network (DDQN), Lu et al. [22] guided the rational selection of machining parameters and improvement of machining efficiency.

Since the DQN-based algorithm learns from the Q value, it is mainly designed for problems with discrete actions. However, the continuous variation of machining parameters poses a challenge in determining the granularity of actions in reinforcement learning. Therefore, various reinforcement learning algorithms that can deal with continuous action space extended by DQN are the focus of researchers in the manufacturing field. Because they are more compatible with the continuous variation process of machining parameters. Wang et al. [23] used asynchronous advantage Actor-Critic (A3C) to reduce the surface defects of complex parts. Huang et al. [24] proposed an integrated control framework for manufacturing systems that optimizes yield by combining Graph Neural Networks (GNN) and reinforcement learning. Xiao et al. [25] used the Actor-Critic framework to improve the energy-saving performance of flexible machining with the integrated meta-enhanced learning model. Zimmerling et al. [26] trained reinforcement learning algorithms under the Actor-Critic architecture to optimize machining parameters for variable workpiece geometries. Due to its adaptability in solving complex problems, DRL is increasingly used to solve multi-objective optimization problems. Li et al. [3,27] optimized machining parameters by improving TD3 to improve machining efficiency and reduce surface roughness or cutting energy. Han et al. [28] proposed a novel differential evolutionary algorithm to solve multi-objective optimization problems combined with reinforcement learning. Li et al. [29] proposed an end-to-end framework for optimization by building multiple sub-neural networks to solve different optimization objectives combined with DRL.

However, existing estimation models and DRL methods often have the following limitations:

1) To reduce the cost of existing DRL modeling, the simulated machining environment is often simplified. As a result, information such as cutting forces during machining is often ignored, and the optimization model does not reflect practical machining processes.

2) Most existing research on DRL is aimed at optimizing a single task. However, it is necessary to optimize multiple objectives collaboratively in actual machining. In addition, existing multi-objective optimization research often relies on multiple independent estimation models, neglecting the interactions between objective factors when solving optimization problems.

3) Insufficient attention is paid to the correlation relationship between state parameters when solving the parameter optimization problem using DRL. This leads to unstable model results when optimizing parameters.

3. Proposed framework

3.1. Problem definition

In batch machining systems, the machining parameters, cutting forces, and machining quality and efficiency of different batches change regularly in a time series manner. The multi-task time series estimation model considers the time-varying nature of milling systems, machining parameters, cutting forces, and their correlation with various machining indicators including machining quality and efficiency indicators. After machining a batch of workpieces, the efficiency indicators and quality indicators of each machined feature are recorded and measured. The subsequent set of blanks with the same specifications are machined on the same table using different combinations of machining parameters. The same recordings and measurements are taken after the machining process is complete.

During machining processes, the combination of machining parameters $para = [n, f, ap]$ typically includes spindle speed n, feed f, and depth

of cut ap. The cutting force $force = [FX, FY, FZ]$ generated is divided into three directions: X-axis, Y-axis, and Z-axis. The cutting force signals in each direction throughout the milling process undergo feature extraction. The extracted cutting force features still have a high feature dimension, which requires high computational cost. Therefore, in order to reduce the computational cost as much as possible without reducing the accuracy of the model, the principal component analysis (PCA) [30] is applied to simplify the feature complexity of the extracted features. The process of dimension reduction is based on the importance of cutting force features relative to the estimated target in the estimation model construction process, and the PCA dimensionality reduction is performed by using the correlation between the features. And Ψ time-domain features are extracted for each direction of cutting force to reflect the overall situation of cutting force under the current machining batch. The cutting force features obtained can be represented by $FX = \{FX^1, ..., FX^\Psi\}$, $FY = \{FY^1, ..., FY^\Psi\}$, and $FZ = \{FZ^1, ..., FZ^\Psi\}$. At the current moment t, the data feature $x_t = [para_t, force_t]$ is constituted by the combination of machining parameters and cutting force features of the workpiece being machined.

The combination of production time series data features for workpieces with a total production number P is labeled as $X = \{x_1, x_2, ..., x_t, ..., x_P\}$. The production time series machining indicators are labeled as $Y = \{y_1, y_2, ..., y_t, ..., y_P\}$. After machining each workpiece, the set of machining quality indicator values for multiple machining features $y^A = \{y^{A,1}, ..., y^{A,q}\}$ and efficiency indicators $y^T = \{y^{T,1}, ..., y^{T,p}\}$ of the current workpiece are measured and recorded. The machining quality indicators for the i-th feature is denoted by $y^{A,i} = \{y^{A,i(1)}, y^{A,i(2)}, ..., y^{A,i(o^i)}\} (i = 1, 2, ..., q)$, where o^i represents the total number of machining quality indicators for the i-th feature and q represents the number of machining features. The j-th machining efficiency indicator is denoted as $y^{T,j} (j = 1, ..., p)$, where p is the number of machining efficiency indicators. The machining quality indicators of current batch, combined with the machining efficiency indicators, comprise the machining indicators $y = [y^A, y^T]$.

To construct the time series estimation model, divide the time series data into sliding time windows. Use e samples to form a time series window and slide it by one step to obtain the data. Designating the starting point of the time series window as $b (b \geq 1)$, a window time series data input feature set of length e is defined as $X_e = \{x_b, x_{b+1}, ..., x_{b+e-1}\}$. To implement time series estimation and machining parameter optimization, the dataset for machining indicators takes a window of data, denoted as $Y_e = \{y_{b+1}, y_{b+2}, ..., y_{b+e}\}$, one step back from the input data features.

The proposed multi-task time series estimation model consists of a time window data $X_{e(t)} = \{x_{t-e+1}, x_{t-e+2}, ..., x_t\} (t - e \geq 0)$ consisting of current workpiece data features x_t and historical data features as input data. Proposed estimation model estimates machining indicators $y_{t+1} = [y^A_{t+1}, y^T_{t+1}]$ for the next workpiece after the time window. Eq. (1) expresses the multi-task time series estimation model. It accurately describes the mapping relationship between time series data features and machining indicators of the future workpiece. This allows for the estimation of machining quality and efficiency indicators, as well as optimization for workpiece machining parameters :

$$y_{t+1} = f_R(X_{e(t)}; \Theta) \tag{1}$$

where f_R represents the multi-task time series estimation model for machining indicators; Θ denotes its set of parameters.

The machining parameter optimization problem is primarily constrained by the quality requirements of the machine tool, milling tool, workpiece size, and machining time. The range of machining parameter values serves as the main constraint for the optimization problem and can be expressed as follows:

$$G(n, f, ap) = \begin{cases} n_{\min} \leq n \leq n_{\max} \\ f_{\min} \leq f \leq f_{\max} \\ ap_{\min} \leq ap \leq ap_{\max} \end{cases} \tag{2}$$

where $G(n, f, ap)$ represents the constraints of the machining parameter optimization problem; the subscript min represents the minimum value that the machining parameters can take, and the subscript max represents the maximum value that the machining parameters can take. The specific values are mainly determined by the machining manual and the actual performance of the machine tool.

Since the reasonable adjustment of milling machining parameters can significantly improve machining quality and efficiency. A multi-objective optimization model with comprehensive consideration of machining quality and efficiency is established with machining parameters as decision variables. The model is used to get the optimal machining parameters of the future workpieces in synergy with the machining indicators of the machining quality and efficiency of the corresponding workpieces optimized. Combining Eq. (1) and Eq. (2), the specific multi-objective optimization and machining parameter optimization task can be expressed as:

$$\begin{cases} para_{best} = argmin(O(n_{t+1}, f_{t+1}, ap_{t+1}) s.t. G(n, f, ap)) \\ O(n_{t+1}, f_{t+1}, ap_{t+1}) = \sum_{i=1}^{q} (|y^{A,i}_{t+1}(n_{t+1}, f_{t+1}, ap_{t+1}) - y^{A,i}_{ideal}|) \\ \qquad + \sum_{j=1}^{p} (|y^{T,j}_{t+1}(n_{t+1}, f_{t+1}, ap_{t+1}) - y^{T,j}_{ideal}|) \\ s.t. y^{A,i}_{\max} \geq y^{A,i}_{t+1}(n_{t+1}, f_{t+1}, ap_{t+1}) \geq y^{A,i}_{\min}, y^{T,j}_{\max} \geq y^{A,j}_{t+1}(n_{t+1}, f_{t+1}, ap_{t+1}) \geq y^{T,j}_{\min} \end{cases} \tag{3}$$

where $O(S_{t+1}, f_{t+1}, ap_{t+1})$ represents the function of multi-objective optimization; $y^{A,i}_{ideal}$ represents the ideal machining quality indicator value of the i-th machining feature, and $y^{T,j}_{ideal}$ represents the ideal machining efficiency indicator value of the j-th workpiece; $y^{A,i}_{\max}$ and $y^{A,i}_{\min}$ represent the maximum and minimum values of the quality indicator of the i-th machining feature; $y^{T,j}_{\max}$ and $y^{T,j}_{\min}$ represent the maximum and minimum values of the j-th workpiece machining efficiency indicator, respectively. The dimensions of different machining indicators are different, in order to unify the calculation and improve the coupling accuracy, all machining indicators data are normalized to between 0 and 1.

3.2. Framework for optimizing machining parameters in the multi-task time series reinforcement learning

The MTE-BDRL framework consists of three parts: the acquisition and construction of batch machining workpiece time series data, construction of the multi-task time series estimation model for multi-objective optimization, and construction of the DRL machining parameter optimization method, as shown in Fig. 2. The three parts are explained as follows:

The acquisition and construction of batch machining workpiece time series data.

Obtaining time series data from a batch milling machining system. The obtained small sample time series data is enhanced to form an enhanced time series dataset [31]. Then the data is divided using the time window, and the data feature characterized by machining parameters and cutting force features of each window are obtained as input data. The multiple machining indicators, including the machining quality and efficiency of each window, are the output objects that are estimated and optimized.

Construction of the multi-task time series estimation model for multi-objective optimization.

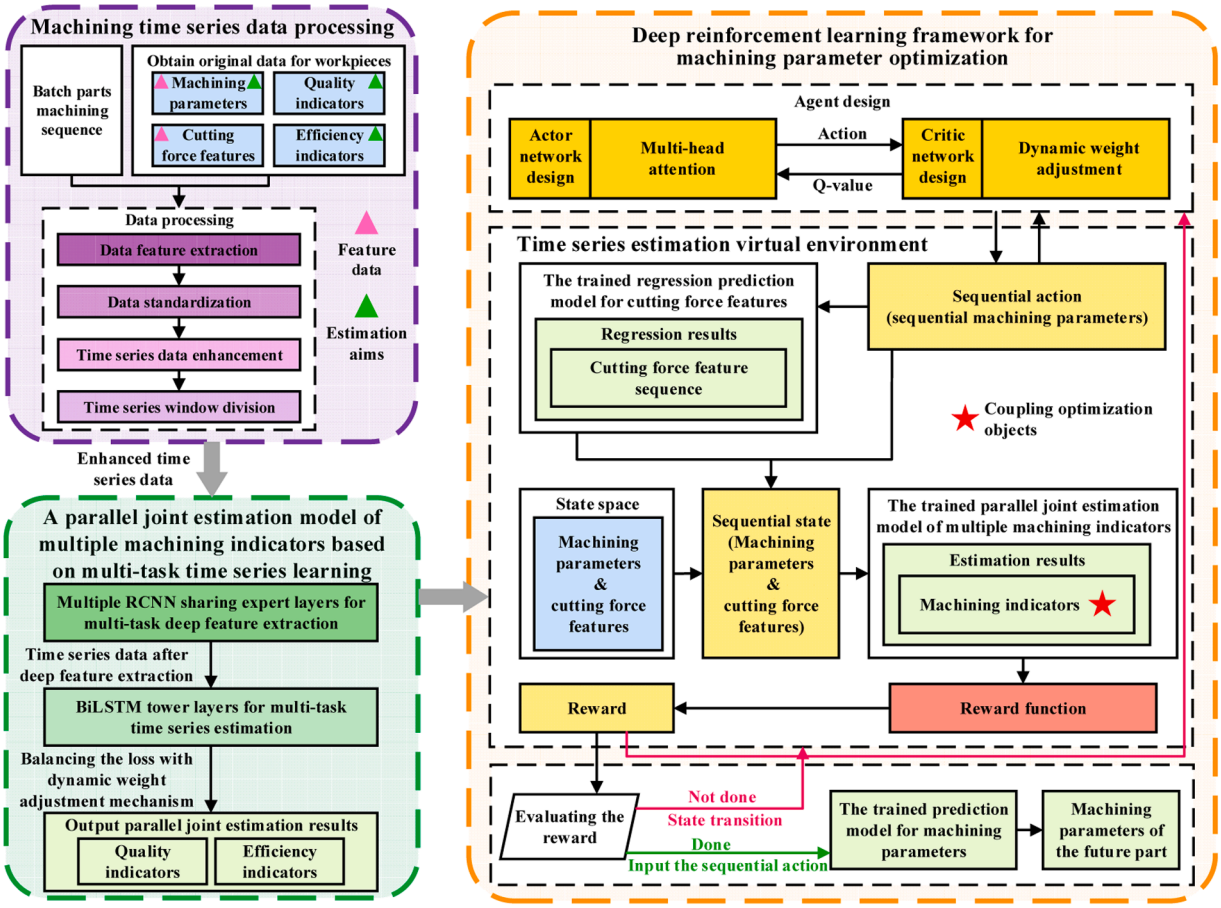

Fig. 2. Framework design of MTE-BDRL.

A multi-task time series estimation model based on multi-gate mixture-of-experts (MMoE) [32] for machining quality and efficiency is designed. Its expert sharing layer is built using Residual Convolutional Neural Networks (RCNN) [33] to extract the deep mathematical relationship between time series data. And its tower layer is made up of Bi-directional Long Short-Term Memory (BiLSTM) [34] to estimate the multiple machining indicators of the future workpieces in parallel. On this basis, an optimization objective function with quality and efficiency as the coupling indicators is constructed.

Construction of the DRL machining parameter optimization method.

An improvement DRL method for collaborative improvement of machining quality and efficiency is constructed to optimize machining parameters of milling systems. By designing a virtual environment supported by the state space and the multi-task time series estimation model with low testing error for machining quality and efficiency, the authenticity of the virtual environment is ensured. In the design of the agent, multi-head attention mechanism is integrated into the actor network to improve the quality of parameter optimization policy, and DWA mechanism is integrated in the loss calculation stage of the critic network to improve the stability of model training.

When the designed DRL framework is executing, the multi-task time series estimation model takes the sequential state as input. Where the output of the estimation model is the machining quality indicators and efficiency indicators. By estimating multiple machining indicators and inputting them into the reward function, the reward value of the corresponding action is calculated. Then the reward value is evaluated whether is close enough to the ideal reward value. If it meets the requirement (done), the action is input into the time series prediction model for the machining parameters to recommend suitable machining parameters for the next workpiece. If the requirement is not met (not done), continue to execute the Markov decision process (MDP) for state transition and search for better optimization parameters.

There are two points to note in the method proposed in this paper:

1) The proposed framework in this paper is universal for batch milling machining, which can provide a reference process for machining parameter optimization based on multiple objectives. Batch milling machining of different workpieces can be learned based on their sample data, resulting in superior model performance. This paper uses thin-walled parts that are difficult-to-machine as an example to demonstrate the effectiveness of the proposed framework, which can be used to optimize machining parameters for other workpieces.

2) This paper explores the value of historical data in optimizing process parameters based on the changing patterns of production machining data and the changing relationships of historical data. Historical data reflects physical mechanisms and is influenced by many physical factors, including milling forces, time series effect of batch production, etc. Therefore, analyzing data can effectively reflect the actual production and play an important role in optimizing machining parameters.

3.3. A parallel joint estimation model of multiple machining quality and efficiency indicators based on multi-task time series learning

Based on the framework of MMoE, the improved Bi-directional multi-gate mixture of experts time series estimation (BMMoE-TE) for the parallel joint estimation model of multiple machining indicators is proposed. This section only describes the estimation model of machining indicators, and its flowchart is shown in Fig. 3. It takes machining

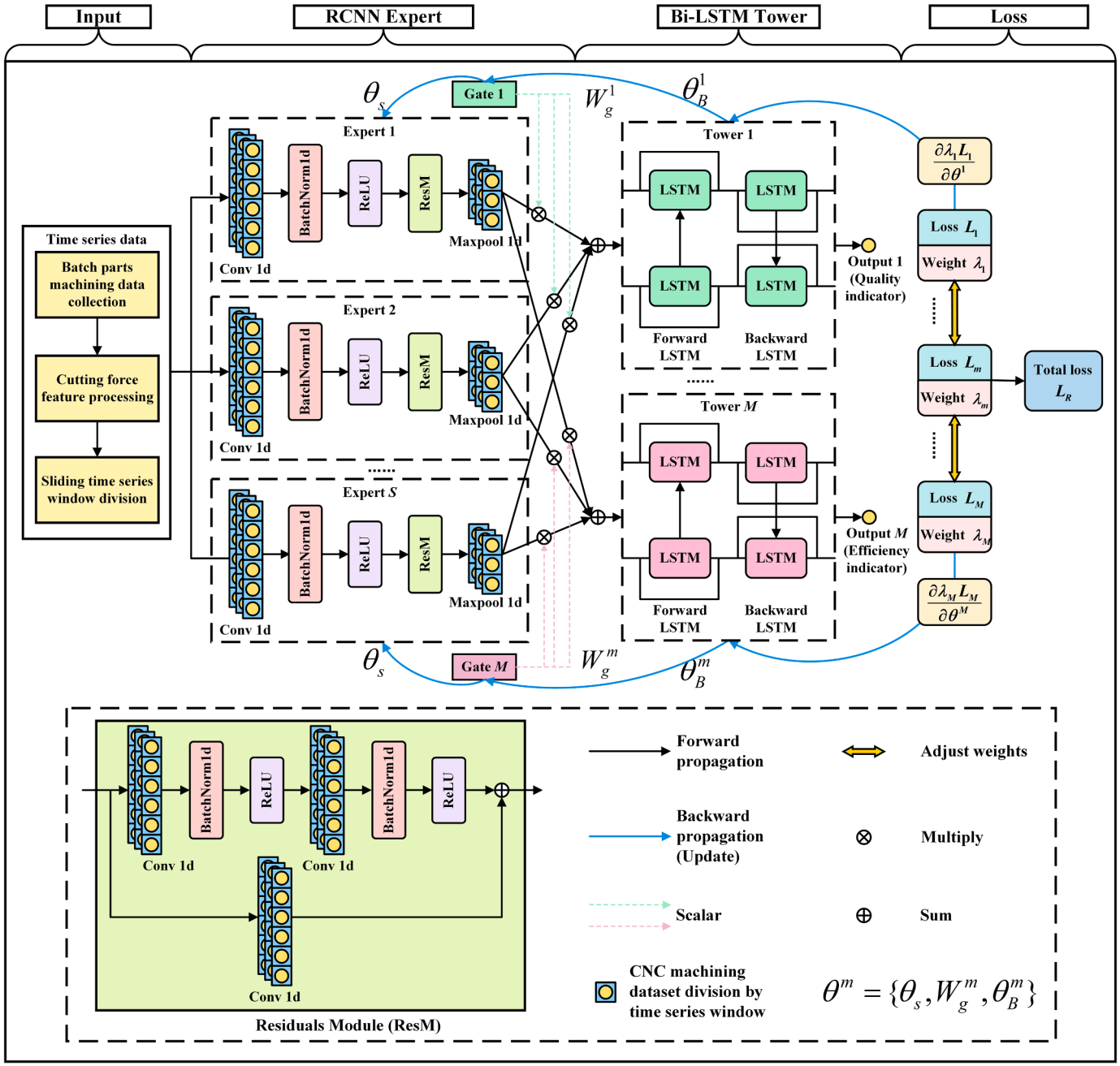

Fig. 3. The proposed BMMoE-TE flowchart for the joint estimation of machining quality and efficiency indicators.

indicators including quality indicators and efficiency indicators as the multi-task estimation aims. The BMMoE-TE model is mainly composed of two modules: the expert sharing layer and the tower layer. In the expert sharing layer of the proposed model, the RCNN is used to jointly extract deep data features and time series features from machining parameters and cutting force data. The tower layer is established corresponding to each task. And BiLSTM is used to perform time series estimation on the fusion feature. To obtain accurate multi-task estimation results, the DWA mechanism is used to assign appropriate weights to each task when training the model. Machining parameters can also be used to construct similar time series analysis, based on optimized machining parameters through historical data, to achieve optimization of future workpiece machining parameters.

3.3.1. *Expert sharing layer network in multiple RCNN*

Based on the characteristics of the short-circuit connection hypothesis, the residual network is easy to train and has fast convergence. It can synchronously extract the data structure and time series features from the time window data. After the data in a time window is processed by RCNN, the length of the time window and the features of the data will change due to feature extraction and transformation. The basic unit is the residual block shown in Eq. (4), and multiple residual blocks are connected to form a deep residual network.

$$\begin{cases} X_{e(t)}^{l+1} = ReLU(C(X_{e(t)}^{l}) + h(X_{e(t)}^{l})) \\ X_{e'(t)} = R(X_{e(t)}, \theta_s) \end{cases} \tag{4}$$

where t represents the current batch, $X_{e(t)}^{l}$ and $X_{e(t)}^{l+1}$ represent the input and output of the l-th residual block; C represents the convolution-pooling-activation operation in the original line; $h(x_t^l) = x_t^l$ represents the identity mapping, which keeps the number of data channels in the

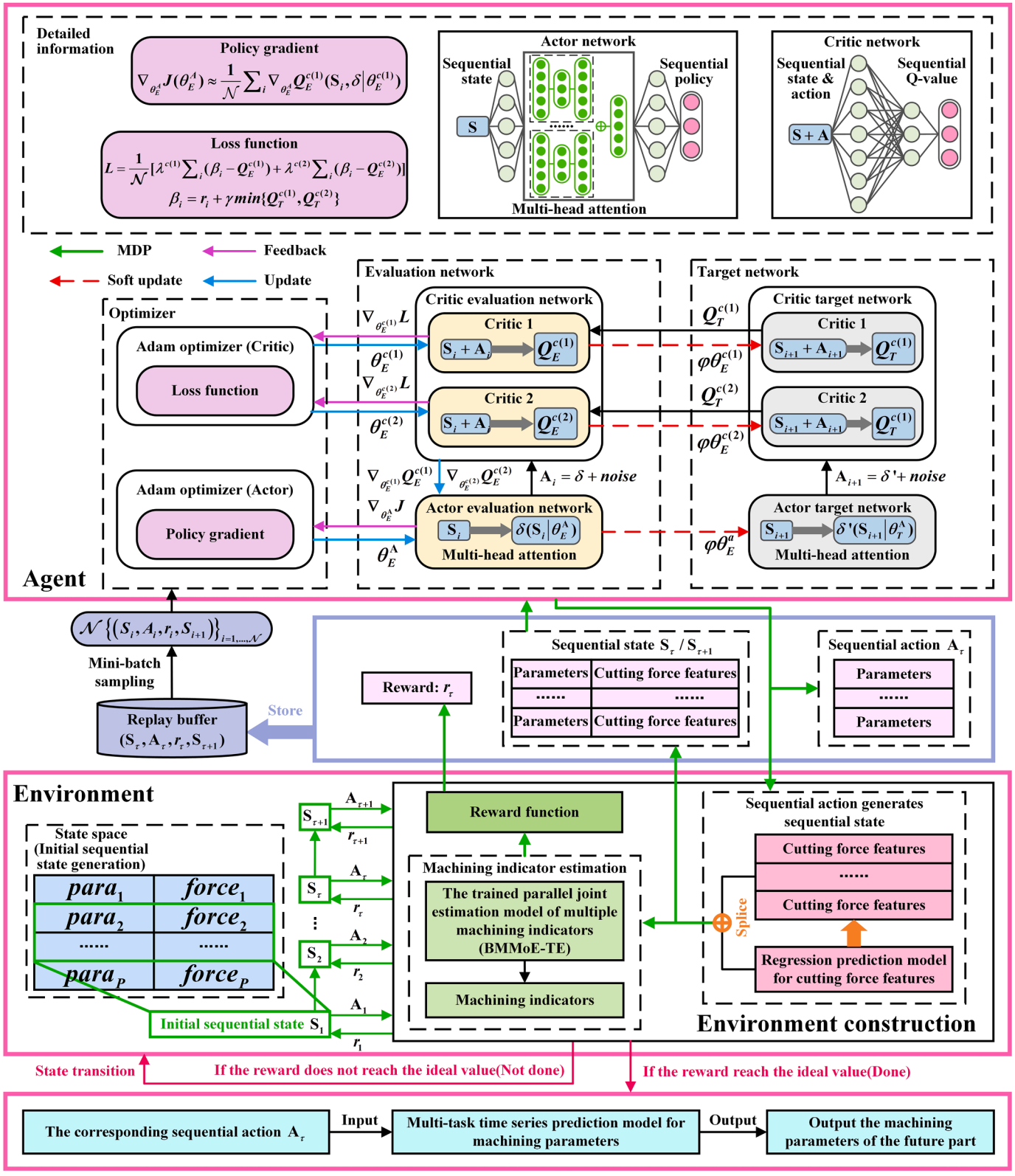

Fig. 4. The structure of MATD3 for optimizing milling machining parameters.

current residual block unchanged. R represents the residual convolution operation, and θ_s represents the parameters of the expert sharing layer; e' represents the length of the time window after processed by residual convolution, and $X_{e'(t)}$ represents the data of a time window after processed by residual convolution.

The features obtained by the underlying shared network $F^m(X_{e(t)}) = \sum_{i=1}^{S} g^m(X_{e(t)})_i (X_{e'(t)})_i$ are expressed as the weighted sum of the features extracted by the expert layer and their corresponding weights. Where $g^m(X_{e(t)}) = softmax(W_g^m X_{e(t)})$ represents the gated network in the expert sharing layer, $W_g^m \in \mathbb{R}^{S\times d}$ is the trainable matrix, S is the number of experts, d is the feature dimension, and m means for *m-th* task. The role of the gate network is to assign weights to expert features through the Softmax activation layer for specific tasks.

3.3.2. Tower layer joint estimation network based on BiLSTM for multiple machining indictors

The BiLSTM as the estimation tower layer for time series machining data is designed to estimate the time series features from the expert sharing layer adjusted by the gate network, as shown in Eq. (5):

$$\widetilde{\varpi_{t+1}^m} = BiLSTM^m(F^m(X_{e(t)}), \theta_B^m) \tag{5}$$

where $\widetilde{\varpi_{t+1}^m}$ represents the estimated output of a single BiLSTM for the next workpiece, that is, the *m-th* estimated value of the output from the multi-task time series estimation model; $BiLSTM^m(\cdot)$ represents the *m-th* BiLSTM transform; θ_B^m represents parameters of the *m-th* tower layer. At the same time, the parameter set of the *m-th* task is $\theta^m = \{\theta_s, W_g^m, \theta_B^m\}$; The total parameter set of the multi-task time series estimation model is $\Theta = \{\theta^1, ..., \theta^M\}$, M is the total number of tasks.

The trained BMMoE-TE model is used to support the virtual environment of the proposed DRL framework, with the state as the input, and the estimated value of the machining indicators of the corresponding future workpiece is output. According to Eq. (4) and Eq. (5), the input $X_{e(t)}$ of the BMMoE-TE model is in sequence form, so the state and the action of the proposed DRL framework are also in sequence form when executing MDP.

The DWA is used to learn the weight of each task λ_m according to the corresponding loss. This is aiming to adjust the adaptive weights among multi-task simply and effectively. By summing the weighted loss of each task, the total loss of the multi-task estimation model with balanced loss weights is finally defined as L_R, which is shown in Eq. (6):

$$L_R = \sum_{m=1}^{M} \lambda_m \left\| \varpi_{t+1}^m - \widetilde{\varpi_{t+1}^m} \right\|^2 \tag{6}$$

where ϖ_{t+1}^m represents the actual observation of the *m-th* estimation task.

In the process of updating and optimizing the multi-task time series estimation model, Adam optimizer and small batch gradient descent are used. the parameter updating of the *m-th* task is expressed as:

$$\theta^m \leftarrow \theta^m - \vartheta^R \left(\frac{\partial \lambda_m \left\| \varpi_{t+1}^m - \widetilde{\varpi_{t+1}^m} \right\|^2}{\partial \theta^m} \right) \tag{7}$$

where ϑ^R represents the learning rate of the multi-task time series estimation model.

Choose R-square (R^2), mean absolute error (MAE) and root mean square error (RMSE) as the performance evaluation metrics of the multi-task time series estimation model. All of them can reflect the estimation accuracy of the model [35]. The pseudocode of the multi-task time series estimation model is shown in Appendix A1–1.

3.4. Machining parameter optimization algorithm based on deep reinforcement learning

3.4.1. The MATD3 design for optimizing milling machining parameters

The DRL framework based on multi-head attention TD3 (MATD3) is designed. It optimizes machining parameters based on multi-objective optimization requirements of machining quality and efficiency, and the structure of MATD3 is shown in Fig. 4. MATD3 includes an improved agent, a virtual environment and a replay buffer. The agent can continuously learn and generate machining parameter optimization policy in MDP. The virtual environment generates and updates the optimized machining parameters through updating the policy. And a series of experience generated during the interaction between the two is stored in the replay buffer.

The agent consists of a multi-head attention-improved actor and two critics. Both the actor and the critic are composed of an evaluation network and a target network. The virtual environment is mainly supported by the trained BMMoE-TE model for multiple machining quality and efficiency indicators. To adapt the trained BMMoE-TE model, the state generated by the virtual environment is characterized by the sequence data of multiple workpieces, and the generated actions are also the execution sequences of multiple workpieces. The trained BMMoE-TE model takes the sequential state as the input, and the multiple machining indicators of the optimized workpiece as the output. The machining indicators serve as collaborative optimization objectives, and the values are utilized to calculate rewards through the reward function, which evaluates the sequential action performed by the agent. Furthermore, the experience generated during operation, which includes the current sequential state, sequential action, rewards, and the next sequential state, are packaged in tuples and stored in the reply buffer with the capacity of $\mathscr{B}$. Experiences are selected randomly in mini-batches of size $\mathscr{N}$ using the experience replay mechanism for training and updating the actor and critic networks.

The designed agent performs cyclic iteration through the change of environmental state and the change of reward value corresponding to the action, and generates multiple optimized machining parameters. The virtual environment updates the state through the multi-task parallel joint time series estimation model based on the repeatedly generated machining parameters, and calculates the corresponding rewards to repeatedly evaluate different machining parameters. Then the sequential machining parameters for the maximum reward are found and input into the machining parameter prediction model to obtain the future workpiece machining parameters, which is the output object of MATD3. The machining parameter prediction model is constructed in the similar way as the BMMoE-TE model in Section 3.3, making sequential machining parameters as input and the future batch machining parameters as multi-task prediction aims.

The sequential state, sequential action, reward function and multi-head attention mechanism in MATD3 are defined as follows:

1) Sequential state definition. The state space $STATE = [PARA, FORCE]$ is formed by arranging the data feature. It consists of all P machining parameter combinations $PARA = \{para_1, para_2, ..., para_P\}$ and cutting force features $FORCE = \{force_1, force_2 ..., force_P\}$, in chronological order. To adapt to BMMoE-TE model, the initial sequential state $\boldsymbol{S}_1$ generated from the state space, generating initial action $\boldsymbol{A}_1$, reward r_1, the second sequential state $\boldsymbol{S}_2$, and subsequent elements that make up the Markov chain, should be sequential. Therefore, the data feature is randomly selected from the state space according to the length of the designed series window as the initial sequential state $\boldsymbol{S}_1$. During the operation of MATD3, the state will be transferred and updated from the initial sequential state $\boldsymbol{S}_1$ to the subsequent sequential states $\boldsymbol{S}_2$,..., $\boldsymbol{S}_\tau$, $\boldsymbol{S}_{\tau+1}$ through the agent executing the action. In this way, MATD3 can explore better parameter optimization policy. As demonstrated in Eq. (8), the sequential state $\boldsymbol{S}_\tau$ generated at the subsequent moment τ is formed by combining the new sequence of machining parameter combinations with the cutting force feature sequence.

$$\begin{cases} \boldsymbol{S}_\tau = [PARA_\tau, FORCE_\tau] \\ FORCE_\tau = \Lambda(PARA_\tau) \end{cases} \tag{8}$$

where Λ represents the arithmetic process of regression prediction of cutting force features using combinations of machining parameters.

2) Sequential action definition. Actions are used to determine machining parameters, and form the force in the sequential state to realize the transition and updating of the sequential state. Then, the machining indicators corresponding to the transferred state is estimated, and the support for the subsequent evaluation of the action is provided on the basis. According to the characteristics of continuous variation of machining parameters, continuous action space is designed. The multi-task time series estimation model runs based on the sequence samples in the history window. Therefore, the sequential action $\boldsymbol{A}_\tau$ at moment τ is represented by Eq. (9), with values within the machining parameter constraints shown in Eq. (2). After sequential action $\boldsymbol{A}_\tau$ generates a new sequence of machining parameters, a corresponding sequence of cutting force features is obtained through regression prediction. The two sequences are then spliced together to generate a new sequential state $s_{\tau+1}$. Although the form of $\boldsymbol{A}_\tau$ is a sequence of machining parameters with window length e, its role is to guide the current state $\boldsymbol{S}_\tau$ to perform state transition and generate the new state $\boldsymbol{S}_{\tau+1}$. On this basis, when the reward value r_τ obtained by the agent action $\boldsymbol{A}_\tau$ reaches the standard, $\boldsymbol{A}_\tau$ is input into the machining parameter estimation model, and the machining parameter $para_{t+1}$ of the future workpiece outside the sequence window is obtained as the recommended values of the machining parameter.

$$\boldsymbol{A}_\tau = \begin{bmatrix} para_{t-e+1} \\ para_{t-e+2} \\ \cdots \\ para_t \end{bmatrix}_\tau \tag{9}$$

3) Reward function definition. Based on the estimated machining indicators, the reward function for evaluating the action is constructed. According to the BMMoE-TE estimation results shown in Eq. (1), for the future workpiece outside the sequence window at the moment τ, the machining parameter $para_{t+1,\tau}$ is the optimized aim. And multivariate machining indicator $y_{t+1,\tau} = [y^{A,i}_{t+1,\tau}, y^{T,j}_{t+1,\tau}]$ is the optimization objective. Under the constraint condition of Eq. (2), the coupling improvement of machining quality and efficiency and the optimization of machining parameters are carried out with Eq. (3) as the aim. Since the optimization process is implemented by the agent executing the sequential action and interacting with the environment. The reward function represented by Eq. (10) is defined. It uses the estimated multivariate machining indicator to calculate the reward value and evaluate the sequential action executed by the agent.

$$\begin{cases} r_\tau(a_\tau|s_\tau) = [\sum_{i=1}^{q}(1 - |y^{A,i}_{t+1,\tau} - y^{A,i}_{ideal}|) + \sum_{j=1}^{p}(1 - |y^{T,j}_{t+1,\tau} - y^{T,j}_{ideal}|)]^2 \\ r_{ideal} = [q+p]^2 \end{cases} \tag{10}$$

where $y^{A,i}_{t+1,\tau}$ and $y^{T,j}_{t+1,\tau}$ represent quality and efficiency indicators estimates for various machining features of future the batch. These estimates are obtained after feeding the new sequential state produced by the moment τ intelligences executing sequential action into the multi-task time series estimation model. To prevent features with small magnitudes from being ignored during the training process, the data is normalized to a range of 0 to 1. Therefore, the difference between the results of the multi-task time series estimation and the ideal value of the corresponding indicator are calculated. Then the absolute value of the interpolated value is subtracted from 1. The results of the calculations under the correspondence of each optimization task are then summed up. And the results are then squared to reflect significant differences between them. r_{ideal} is the ideal reward value when $y^{A,i}_{t+1,\tau}$ and $y^{T,j}_{t+1,\tau}$ reach the ideal value $y^{A,i}_{ideal}$ and $y^{T,j}_{ideal}$. When the optimization policy converges towards increasing rewards, the agent will optimize machining parameters that synergistically optimize machining quality and efficiency. Finally, when the reward is close enough to the r_{ideal}, the machining parameters of the corresponding action are obtained.

4) Multi-head attention mechanism. The process of searching policy is realized by the actor network. By adding the multi-head attention mechanism between the linear layers of the actor network, the computation process of multi-head attention can be denoted by $MA(\boldsymbol{S}_\tau \otimes \vec{\&} \parallel \boldsymbol{S}^{MA}_\tau)$. $\boldsymbol{S}_\tau$ is input into the actor network at moment τ, and $\boldsymbol{S}^{MA}_\tau$ represents the state output after multi-head attention computation. The process of computing and rationally adapting the association relationship to the state data is denoted by $\otimes \& \parallel$. This is achieved using self-attention and splicing the results of multiple self-attention computations to form the multi-head attention output [36].

3.4.2. Virtual environment construction

The constructed virtual environment here comprises two main parts: the state space, which is composed of time-ordered state space, and the BMMoE-TE model. The state space can generate time series machining parameters and cutting force features sequence data to form the initial sequential state $\boldsymbol{S}_1$ for MDP iteration. In MDP, the agent generates the sequential action $\boldsymbol{A}_\tau$, for the corresponding moment based on the sequential state $\boldsymbol{S}_{\tau+1}$ at moment τ. The BMMoE-TE model is utilized to evaluate the sequential action $\boldsymbol{A}_\tau$ performed by the agent and estimate machining indicators for the upcoming batch. This is based on the new sequential state generated $\boldsymbol{S}_{\tau+1}$ by the agent. The reward value r_τ is then calculated using Eq. (10). The above process is shown by the green arrow in Fig. 4. The replay buffer stores $(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau, r_\tau, \boldsymbol{S}_{\tau+1})$ and samples it with a mini-batch of $\mathcal{N}$. The mini-batch sampled data $\mathcal{N}\{(\boldsymbol{S}_i, \boldsymbol{A}_i, r_i, \boldsymbol{S}_{i+1})\}_{i=1,\ldots,\mathcal{N}}$ is then used to train and update the networks, which is shown by blue arrows, purple arrows, and red dotted arrows in Fig. 4.

3.4.3. The agent construction and execution

The agent construction and execution process flow of the MATD3 includes the following steps:

Step 1: The actor evaluation network with parameter $\theta^{\boldsymbol{A}}_E$ is designed by taking a sequential state $\boldsymbol{S}_\tau$ as input and produces a sequential action $\boldsymbol{A}_\tau = \delta(MA(\boldsymbol{S}_\tau \otimes \vec{\&} \parallel \boldsymbol{S}^{MA}_\tau)|\theta^{\boldsymbol{A}}_E) + \xi_\tau$, using the multi-head attention mechanism. Where δ represents the current policy of the evaluation network, and ξ_τ represents ornstein-uhlenbeck (OU) noise [37]. The OU process generates timing-related noise, which causes the agent to explore more in the direction of greater cumulative rewards. This improves the efficiency of timing-related exploration by the agent.

Step 2: The two critic evaluation networks with parameters $\theta^{c(1)}_E$ and $\theta^{c(2)}_E$ are designed with the input sequential state $\boldsymbol{S}_\tau$ and sequential action $\boldsymbol{A}_\tau$ to calculate the Q-value. It can evaluate the sequential action $\boldsymbol{A}_\tau$ selected by the actor evaluation network. Because state $\boldsymbol{S}_\tau$ and action $\boldsymbol{A}_\tau$ are sequential data, the Q-value computed by critical evaluation networks are also sequential data. To update the network, the Q-values within the sequence are averaged to obtain the corresponding $Q^{c(1)}_E(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau|\theta^{c(1)}_E)$ and $Q^{c(2)}_E(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau|\theta^{c(2)}_E)$ as Eq. (11):

$$\begin{cases} Q^{c(1)}_E(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau|\theta^{c(1)}_E) = \dfrac{\sum_{z=t-e+1}^{e} Q^{c(1)}_{E,z}}{e} \\ Q^{c(2)}_E(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau|\theta^{c(2)}_E) = \dfrac{\sum_{z=t-e+1}^{e} Q^{c(2)}_{E,z}}{e} \end{cases} \tag{11}$$

where $Q^{c(1)}_{E,z}$ and $Q^{c(2)}_{E,z}$ represent the elements inside the Q-value of the sequence output by the two critic evaluation networks, respectively.

Step 3: The actor target network outputs the sequential action $\boldsymbol{A}_{\tau+1}$ for the next sequential state. To prevent overfitting in the DPG method,

reducing estimation bias and abruptness in the computation process through the addition of truncated normally distributed noise (i.e., $N(\cdot)$). The sequential action constructed by the actor target network is then cropped $clip(\cdot,\cdot,\cdot)$ to ensure that the actions satisfy the machining parameter constraint range conditions. The process is represented as $\boldsymbol{A}_{\tau+1} = \delta'(MA(\boldsymbol{S}_{\tau+1} \overrightarrow{\otimes \&} \| \boldsymbol{S}_{\tau+1}^{MA}) | \theta_T^{\boldsymbol{A}}) + clip(N(0,\mu), -c, c)$, where δ' represents the current policy of the target network, $-c$ and c represent the upper and lower cropping limits.

Step 4: The two critic target networks with parameters $\theta_T^{c(1)}$ and $\theta_T^{c(2)}$ use the next moment sequential state sequential $s_{\tau+1}$ and next sequential action $a_{\tau+1}$ as inputs. And they obtain the target Q-values $Q_T^{c(1)}(\boldsymbol{S}_{\tau+1}, \boldsymbol{A}_{\tau+1} | \theta_T^{c(1)})$ and $Q_T^{c(2)}(\boldsymbol{S}_{\tau+1}, \boldsymbol{A}_{\tau+1} | \theta_T^{c(2)})$ by averaging the sequence target Q-values internally in the same way as in Eq. (11).

Step 5: Based on the above process, the Temporal-Difference target $\beta_\tau = r_\tau + \gamma \min\{Q_T^{c(1)}(\boldsymbol{S}_{\tau+1}, \boldsymbol{A}_{\tau+1} | \theta_T^{c(1)}), Q_T^{c(2)}(\boldsymbol{S}_{\tau+1}, \boldsymbol{A}_{\tau+1} | \theta_T^{c(2)})\}$ is computed using the smaller of the two target Q-values. This helps to mitigate the network's overestimation of Q-values. Where $\gamma \in [0,1]$ represents the discount factor, which reflects the agent's preference for present and future rewards. It can be determined through the search method.

Step 6: Using Eq. (12), the time-difference errors ρ_1 and ρ_2 [38] generated by the time-difference target and each of the two critic networks will be calculated.

$$\rho_{1/2} = \beta_\tau - Q_E^{c(1)/c(2)}(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau | \theta_E^{c(1)/c(2)}) \tag{12}$$

Step 7: Return to Step 1 with the new sequential state $\boldsymbol{S}_{\tau+1}$ as the policy computation object. Input it to the actor evaluation network and enter the MDP for the next moment $\tau + 1$. Finally, the operation of MATD3 is completed and the optimized machining parameters are output.

3.4.4. MATD3 loss dynamic weight adjustment mechanism and network updating

The loss and network parameters update in the MATD3 is based on the min-batch data extracted from the replay buffer. Due to possible differences in the stability of the updating of the two critic networks of MATD3, the Q-values output by the two critic networks may have different confidence levels. Since the process of calculating the loss function L may be biased, leading to incorrect direction of the agent's activities. To balance the stability of the two critic networks during updating, the DWA [39] is introduced during the training process. Where the weights are assigned to the losses $\frac{1}{\mathcal{N}}\sum_i {\rho_1}^2$ and $\frac{1}{\mathcal{N}}\sum_i {\rho_2}^2$ of each of the two critic networks to compute the loss function L of the critic network according to Eq. (13):

$$L(\theta_E^{c(1)}, \theta_E^{c(2)}) = \frac{\lambda^{c(1)}}{\mathcal{N}}\sum\nolimits_i {\rho_1}^2 + \frac{\lambda^{c(2)}}{\mathcal{N}}\sum\nolimits_i {\rho_2}^2 \tag{13}$$

where $\lambda^{c(1)}$ and $\lambda^{c(2)}$ represent the weights of the respective losses of the two critic networks. To minimize this loss function, critic evaluation networks update themselves by making L achieve gradient descent with respect to $\theta_E^{c(1)}, \theta_E^{c(2)}$ using Adam optimizer.

The aim of the actor network is to maximize the expectation of cumulative rewards. Its parameters are updated by sampling from the buffer and combining the Q-function evaluated by the critic evaluation network to train and improve the policy gradient. The process can be represented by Eq. (14):

$$\nabla_{\theta_E^{\boldsymbol{A}}} J(\theta_E^{\boldsymbol{A}}) = \frac{1}{\mathcal{N}}\sum\nolimits_i \nabla_{\boldsymbol{A}} Q_E^{c(1)}(\boldsymbol{S}_i, \boldsymbol{A}_i | \theta_E^{c(1)}) \cdot \nabla_{\theta_E^{\boldsymbol{A}}} \delta(\boldsymbol{S}_i | \theta_E^{\boldsymbol{A}}) \tag{14}$$

where $\nabla_{\theta_E^{\boldsymbol{A}}} J(\theta_E^{\boldsymbol{A}})$ represents the gradient of the objective function $J(\theta_E^{\boldsymbol{A}})$ with respect to the parameter $\theta_E^{\boldsymbol{A}}$; $\nabla_{\boldsymbol{A}} Q_E^{c(1)}(\boldsymbol{S}_i, \boldsymbol{A}_i | \theta_E^{c(1)})$ represents the gradient of the Q-value with respect to the sequential action, and $\nabla_{\theta_E^{\boldsymbol{A}}} \delta(\boldsymbol{S}_i | \theta_E^{\boldsymbol{A}})$ represents the gradient of the policy δ with respect to the parameter $\theta_E^{\boldsymbol{A}}$.

The critic evaluation network guides the updating of the actor network, while the target network provides a stabilizing aim for network updating and function fitting. If there is a significant deviation between the critic evaluation network and the target network, residual errors can accumulate during network training. It can result in instability in the critic network training process and divergence in the actor network. Therefore, the critic network is updated for a fixed number of steps before updating the actor network. This improves the stability of the critic network and the quality of the policies produced by the actor network. During the process of updating network parameters, MATD3 utilizes a soft update mode. This means that the target network parameters are periodically updated online using the update coefficients $\varphi (0 < \varphi < 1)$. The process can be expressed as follows:

$$\begin{cases} \theta_T^{c(1)} \leftarrow \varphi \theta_E^{c(1)} + (1-\varphi)\theta_T^{c(1)} \\ \theta_T^{c(2)} \leftarrow \varphi \theta_E^{c(2)} + (1-\varphi)\theta_T^{c(2)} \\ \theta_T^{\boldsymbol{A}} \leftarrow \varphi \theta_E^{\boldsymbol{A}} + (1-\varphi)\theta_T^{\boldsymbol{A}} \end{cases} \tag{15}$$

The pseudocode for MATD3 can be found in Appendix A1–2.

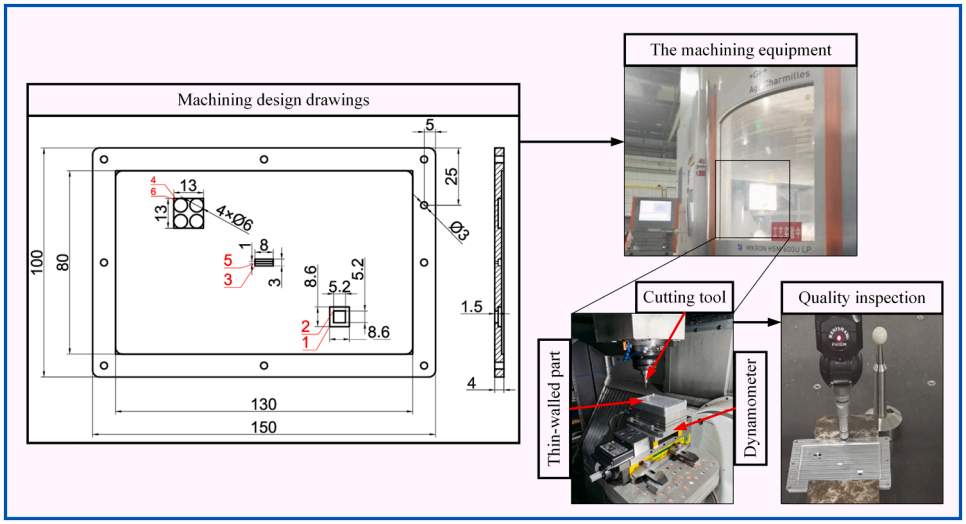

Fig. 5. Key information of the thin-walled part experimental data acquisition process.

4. Experiment verification

4.1. Data description

The shape size and key machining feature information of the thin-walled part and the CNC milling experimental data acquisition process are shown in Fig. 5. The overall length of the thin-walled part is 150 mm, the width is 100 mm, and the thickness is 4 mm. It has six machining features, including 5 types of rectangles and 1 type of circle. The diameter of the circle is 4 mm, the length and width of rectangle 1 are 8.6 mm, the length and width of rectangle 2 are 5.2 mm, the length of rectangle 3 is 8.0 mm and the width is 3.0 mm, the length and width of rectangle 4 are 13.0 mm, and the length of rectangle 5 is 8.0 mm and the width is 1.0 mm. Among them, for the rectangle 5, because its length is the same as rectangle 3, when measuring the dimensional error, only its width is concerned, which is called crack.

The machining experiment of the thin-walled part is carried out on the Mikron HSM600ULP high-speed machining center. For the milling process of each workpiece, each machining feature is milled from the bottom right to the top left in turn. Firstly, the rectangle 1 and the rectangle 2 are milled. Then the cutting tool moves up to the left, and mills the rectangle 3 and the rectangle 5. Finally, the cutting tool is moved to the upper left corner. The rectangle 4 and four circles are milled respectively.

When machining each workpiece, the current three milling machining parameters (i.e., spindle speed, feed, and depth of cut) are recorded. Where the combination of machining parameters of different workpieces is set by the orthogonal experiment. At the same time, cutting force signals of the X-axis, Y-axis, and Z-axis during machining is obtained through a 9527 Ad Kistler dynamometer. In the batch milling experiment, the same machine tool is used to mill the blanks of the same material according to the same size setting. During the machining process, the used cutting tools also maintain the same specifications. In the process of batch milling, 5 cutting tools of the same specifications are used to mill the workpieces, all of which are aluminum alloy flat-end milling cutting tools with the diameter of 1 mm. On this basis, the recommended machining parameters also correspond to the machining of the same cutting tools, which is in line with the actual machining parameters recommended in business scenarios. After each workpiece machining, the milling completion time of all machining features is recorded, and then the dimensional error values are calculated by the design sizes and the actual measured values of the machining features. Where the actual measured values of the machining features are obtained by the 3-coordinate measuring instrument.

The cutting force signals collected by the dynamometer are processed by time-domain feature extraction to form the original cutting force features. After feature extraction, 9-dimensions (mean, maximum, standard deviation, root mean square, peak-to-peak value, skewness index, kurtosis index, peak index, and pulse index) in the three directions of the X-axis, Y-axis, and Z-axis are obtained, that is, a total of 27-dimensional original time-domain features. In order to reduce the computational cost, it is necessary to reduce the dimension of data features to accelerate the convergence of the training model. The results of dimensionality reduction are as shown in Fig. 6. PCA is used to reduce the original 9-dimensional time-domain features in all directions to 3-dimensional, forming a total of 9-dimensional cutting force features. The obtained 9-dimensional features can represent about 99 % of the information representation of the original 27-dimensional cutting force features. Then, the cutting force features and machining parameters are used as the data feature to estimate the machining indicators. Therefore, the importance of the feature after dimension reduction is basically unchanged relative to the estimated machining indicators, which can be used to construct the multi-task time series estimation model.

In summary, machining parameters, cutting force features, machining feature dimensional error values and workpiece machining time together form the CNC machining dataset of thin-walled parts shown in Table 1. The dataset is divided into the data feature and the machining indicator. The machining parameters and cutting force features are composed of data feature as the input of the BMMoE-TE. The dimensional error values of machining features are the machining quality indicators of the workpiece, and the machining time of the workpiece is the machining efficiency indicator. The two together form

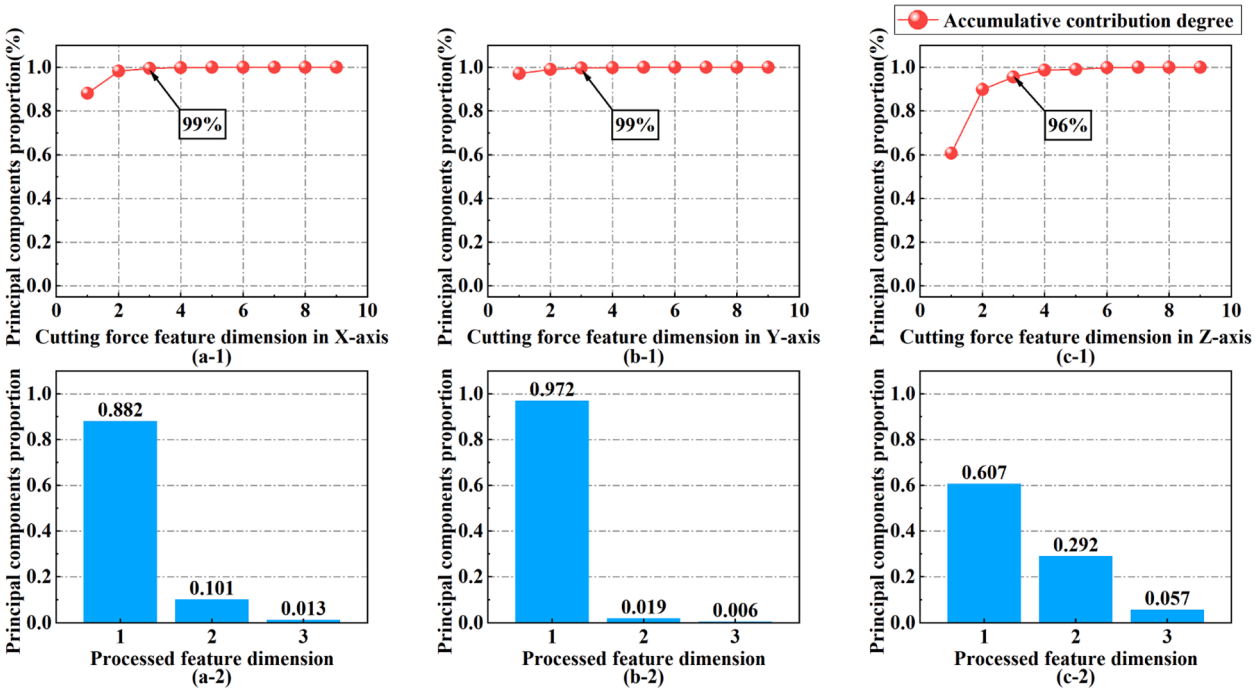

Fig. 6. Analysis of the principal components proportion. (a-1), (b-1) and (c-1) are the proportion of principal components that can be retained when the time-domain features of cutting force in X, Y and Z directions are reduced to the specified dimension. (a-2), (b-2) and (c-2) are the proportion of principal components of each dimension feature after the feature in each direction is reduced to 3 dimensions respectively.

Table 1
The CNC machining datasets of the thin-walled part.

Data feature x	Machining parameter *para*	Cutting force feature FX, FY, FZ	
	Spindle speed (r/min) n Depth of cut (mm) f Feed (mm/min) ap	The 9-dimensional cutting force time-domain features are obtained by PCA processing of the original 27-dimensional time-domain features. $FX^1, FX^2, FX^3, FY^1, FY^2, FY^3, FZ^1, FZ^2, FZ^3$	
Machining indicator y		Machining indicators	
	Machining feature types (mm)	Machining quality indicator y^A	Machining time y^T
	1: Rectangle 1 $y^{A,1}$ (length×width: 8.6 ×8.6)	Rectangle 1 length $y^{A,1(1)}$ as Task 1 Rectangle 1 width $y^{A,1(2)}$ as Task 2	
	2: Rectangle 2 $y^{A,2}$ (length×width: 5.2 ×5.2)	Rectangle 2 length $y^{A,2(1)}$ as Task 3 Rectangle 2 width $y^{A,2(2)}$ as Task 4	
	3: Rectangle 3 $y^{A,3}$ (length×width: 8.0 ×3.0)	Rectangle 3 length $y^{A,3(1)}$ as Task 5 Rectangle 3 width $y^{A,3(2)}$ as Task 6	Workpiece machining time $y^{T,1}$ as Task 11
	4: Rectangle 4 $y^{A,4}$ (length×width: 13.0 ×13.0)	Rectangle 4 length $y^{A,4(1)}$ as Task 7 Rectangle 4 width $y^{A,4(2)}$ as Task 8	
	5: Rectangle 5 $y^{A,5}$ (length×width: 8.0 ×1.0)	Crack size $y^{A,5(1)}$ as Task 9	
	6: Circle $y^{A,6}$ (Diameter: 6.0)	Diameter $y^{A,6(1)}$ as Task 10	

the machining indicator data as the estimated output target of the BMMoE-TE.

In the batch milling scenario, a total of 40 workpieces are machined. The data extracted from these 40 workpieces is used as the original dataset, that is, historical data. The complete time series dataset formed by the combination of milling data feature and machining indicator data is used for systematic training of estimation and optimization models. The actual CNC machining is a small batch manufacturing process, which only produces a small number of data samples. In order to expand the sample number of model training and testing, the Hermite interpolation method [31] is used to enhance the isomorphism of the original dataset, and a time series dataset containing 1950 samples is formed. Fig. 7 shows the distribution of some data before and after isomorphic enhancement, indicating that enhanced data distribution is similar to the original data. This ensures that the constructed model still has the attributes given by the original data, so that the trained model can be applied to the original data. Each of the 1950 sets of data contains the same content as each of the original 40 data. Each data is composed of the data feature and the machining indicator, including machining parameters, cutting force features, machining quality indicators and machining efficiency indicators. In the parallel joint estimation experiment of multiple machining quality and efficiency indicators, 75 % of the time series dataset is randomly selected as the training set and the remaining 25 % is used as the test set. The dataset is divided by sliding with a time series window of length 3, where the sliding step size is 1. The data feature samples of a window including the data feature and the machining indicator data are used as the input of the BMMoE-TE. The output of the BMMoE-TE is 11 estimation tasks, including 10 machining quality indicators and 1 machining efficiency indicator.

The proposed MATD3 in this paper optimizes the machining parameters of the future batch thin-walled parts. The virtual environment of MATD3 mainly consists of state space and the trained BMMoE-TE model. In order to adapt to the BMMoE-TE model, the workpiece data with window length of 3 is randomly selected from the state space composed of 1950 data as the initial sequence state $\boldsymbol{S}_1$. The sequence state includes machining parameters and cutting force features, and the subsequent sequence state is obtained by the initial sequence state $\boldsymbol{S}_1$ through the interaction between the agent and the environment. Finally, through the evaluation of MATD3, the machining parameters of a future batch of workpiece outside the window are optimized. The sequential state is the input of the agent. After inputting the sequential state of moment τ to the agent, the sequential action $\boldsymbol{A}_\tau$ composed of machining parameters with the sequence length of 3 is obtained by calculating the deterministic policy and adding noise. The corresponding cutting force features with the sequence length of 3 are then obtained by regression prediction for $\boldsymbol{A}_\tau$. The sequential action and the cutting force feature sequence are spliced to form the sequential state of the next moment with the sequence length of 3 in MDP. This sequential state can couple and characterize the 1950 milling data feature. Then the machining indicators estimation result for the future batch corresponding to sequential action $\boldsymbol{A}_\tau$ at moment τ is obtained by the estimation model. The reward value is obtained by the reward function shown in Eq. (10) with the estimation results of the machining indicators.

In the optimization process, the ideal values in the reward function and the objective function are set. Combined with actual production experience, historical delivery orders and equipment conditions, the ideal efficiency indicator y^T_{ideal} (i.e., completion time) is set to 30000 s. Since the thin-walled parts studied in this paper are finely machined, the quality indicators should be as small as possible, so the ideal quality

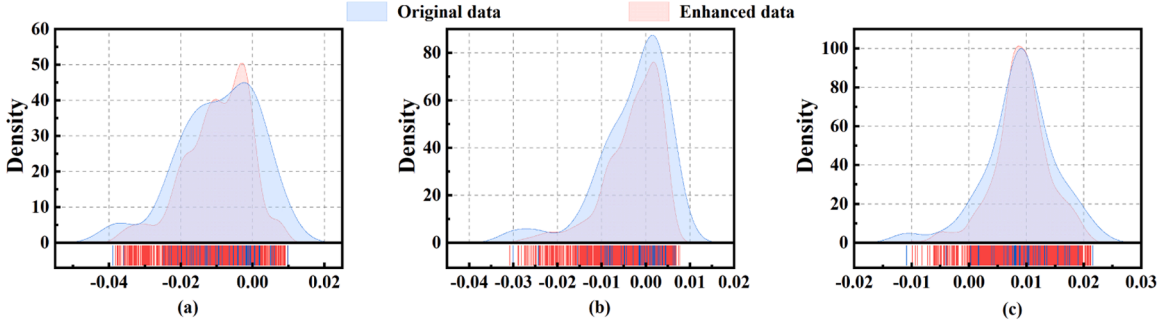

Fig. 7. Comparison of data distribution before and after isomorphism enhancement. (a) Rectangle 1 wide data distribution (b) Rectangle 4-length data distribution (c) Crack size data distribution.

Table 2
Structural parameter settings of the BMMoE-TE model.

Network types	Structure name	Output size
Expert network with RCNN (Experts number: 8 Sequence length: 3)	Input layer (Permute)	(12, 3)
	Conv1d 1 (Kernel size: 2, Stride: 1) (ReLU)	(36, 4)
	Maxpool1d 1 (Kernel size: 2, Stride: 2)	(36, 2)
	Residual block 1 (Kernel size: 3, Stride: 1)	(36, 2)
	Conv1d 2 (Kernel size: 2, Stride: 1) (ReLU)	(18, 3)
	Maxpool1d 2 (Kernel size: 2, Stride: 2)	(18, 1)
	Residual block 2 (Kernel size: 3, Stride: 1)	(18, 1)
	Conv1d 3 (Kernel size: 2, Stride: 1) (ReLU)	(8, 2)
	Maxpool1d 3 (Kernel size: 2, Stride: 2)	(8, 1)
	Residual block 3 (Kernel size: 3, Stride: 1)	(8, 1)
	Output layer (Permute)	(1, 8)
Tower network with BiLSTM (Tasks number: 11 Sequence length: 3)	Input layer	(1, 8)
	BiLSTM layer	(1, 1)
	Output layer	(1, 1)

indicators $y_{ideal}^{A,i}$ (i.e., dimensional error values) are all set to 0.

4.2. Approaches implementation

4.2.1. The proposed model design and hyperparameter determination

The BMMoE-TE and the MATD3 are constructed using the method proposed in Section 3.3 and Section 3.4, which utilizes the dataset extended through data enhancement techniques. Since different numerical combinations of hyperparameters can impact the estimation accuracy of BMMoE-TE and the training process of the MATD3. Therefore, the structures of BMMoE-TE and MATD3 are determined by the combination of random search [40] and grid search [41]. The general grid search is based on experience to artificially give traversal values, but this may lead to some better possible values being ignored. Therefore, the random search is used to determine several specific values for each parameter, and then the grid search is used for traversal tuning. The specific structures of BMMoE-TE for machining indicators and MATD3 are analyzed through numerous experimental results as shown in Table 2 and Table 3, respectively. The structure of the multi-task time series prediction model for machining parameters is similar to the BMMoE-TE model. The detailed structure of MATD3 is shown in Fig. 8.

During the hyperparameter optimization of the BMMoE-TE, several specific values are selected for the hyperparameters by random search method. Then these values are traversed by the grid search method to determine the final hyperparameter combination. The key parameters and adjustment ranges of BMMoE-TE are listed in Table 4. Taking R^2, MAE, and RMSE as the evaluation indicators of hyper-parameter combination, the estimation accuracy of the model under different parameter groups is compared. According to the comparison results shown in Fig. 9, after the training results tend to be stable, when the batch size is 32, the learning rate is 0.001 and the number of experts is 8, the model obtains the best benefit.

The hyperparameter determination method of MATD3 is the same as that of BMMoE-TE. During the tuning phase of the MATD3 model, discount factor, mini-batch size, and the learning rate of actor network and key network are adjusted to select the optimal values. In the tuning phase, the maximum and minimum values of each tuning object are set. The specific values are selected in the given interval by random search for further tuning. Pairwise combinations are made using different values of the learning rate for both the actor and critic networks. Then the grid search method is applied to compare and adjust the discount factor and mini-batch size based on the selected specific values. According to the values of ① ② ③ ④, the learning rate combination is divided into four group, and Table 5 displays the specific model tuning parameter ranges.

Several experiments have shown that the model stabilizes at around 100 episodes. Therefore, the model is set to run at 150 episodes for each parameter combination. The average returns of the last 50 episodes are used as the basis for tuning comparisons among the different subgroups. Based on the comparison results presented in Fig. 10, it is evident that the model with 0.001 learning rate for the actor network, 0.001 learning rate for the critic network, 0.95 discount factor, and a mini-batch of 16, achieves the optimal average return after the training results have stabilized.

4.2.2. Comparison methods and their structural explanations

To compare the performance of the proposed multi-task time series estimation model and the machining parameter optimization method, the comparative experiments are set up for each of the two models as shown in Fig. 11. Before the experimental verification, the expanded historical data is first used to train and test BMMoE-TE and MATD3 and each comparison model to ensure that they all achieve the best performance. The comparison models are explained as follows.

BMMoE-TE comparative experimental design.

Firstly, the proposed BMMoE-TE is compared with single-task time series estimation models to prove the necessity of parallel estimation when estimating multiple machining indicators. The single-task estimation model RCNN-BiLSTM-TE, as shown in Fig. 11, uses RCNN to extract internal information from the data, and uses BiLSTM to estimate the time series of the task. The extended time series historical data of CNC machining of thin-walled parts are input into 11 single-task models,

Table 3
Structural parameter settings of the MATD3 model.

Network types	Structure name	Output size
Actor network (Sequence length: 3)	Input layer	(3, 12)
	Linear layer 1 (Nodes: 256)	(3, 256)
	Multi-head attention layer 1 (Heads: 4)	(3, 256)
	Linear layer 2 (Nodes: 128)	(3, 128)
	Multi-head attention layer 2 (Heads: 4)	(3, 128)
	Linear layer 3 (Nodes: 64)	(3, 64)
	Multi-head attention layer 3 (Heads: 4)	(3, 64)
	Output layer 4 (Nodes: 3)	(3, 3)
Critic network (Sequence length: 3)	Input layer	(3, 15)
	Linear layer 1 (Nodes: 256)	(3, 256)
	Linear layer 2 (Nodes: 128)	(3, 128)
	Linear layer 3 (Nodes: 64)	(3, 64)
	Output layer 4 (Nodes: 1)	(3, 1)

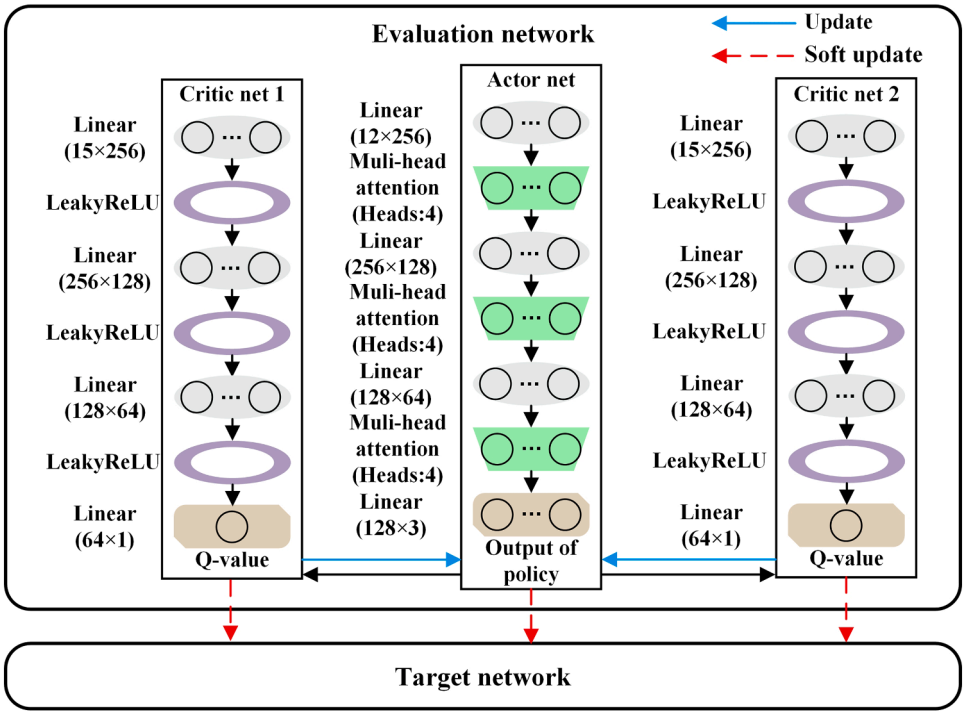

Fig. 8. The detailed structure of MATD3.

Table 4
BMMoE-TE tuning parameter range.

Hyperparameters	Parameter space
Learning rate	{5e−3, 1e−3, 5e−4, 1e−4}
Number of experts	{2, 4, 6, 8}
Batch size	{16, 32, 64}
Optimiser	Adam

and the single-task models of 11 estimation tasks are trained as the comparison models.

Secondly, the different multi-task time series models based on the MMoE framework are compared with the proposed BMMoE-TE model to prove the superiority of the proposed BMMoE-TE structure in multi-task estimation. The 4 multi-task estimation models shown in Fig. 11 are BMMoE-TE (CNN-BiLSTM), gate multi-gate mixture of experts time series estimation (GMMoE-TE (CNN-GRU)), multi-gate mixture of experts time series estimation (MMoE-TE (RCNN-LSTM and RCNN-RNN)). Similar to BMMoE-TE (RCNN-BiLSTM), the comparison model sets the expert shared layer of MMoE as either CNN [42] or RCNN [33]. And the tower structures of MMoE are set as BiLSTM [34], LSTM [43], Recurrent Neural Network (RNN) [44], or Gate Recurrent Unit (GRU) [45] for multi-task time series estimation. The comparison models are trained based on the same dataset, and the structure parameters and hyperparameters of comparison estimation models are determined by random search and grid search.

(2) MATD3 comparative experimental design.

The proposed MATD3 is compared with different deep reinforcement learning models with the aim of demonstrating the advantages of MATD3. Since the proposed MATD3 is a reinforcement learning algorithm based on policy gradient for continuous action space. Therefore, four DRL algorithms, TD3 [38], deep deterministic policy gradient (DDPG) [46], multi-head attention DDPG (MADDPG) and residuals TD3 (RTD3) in Fig. 11 are selected as comparison algorithms. Based on DDPG, MADDPG integrates the multi-head attention mechanism into the linear layers of the actor network. Based on Transformer [47], RTD3 adds a linear layer branch to each multi-head attention module in the actor network of MATD3. This branch outputs the original input directly without using multi-head attention after the linear layer calculation. At the end of the branch line, the linear output is added to the original output and used as the input for the next module. The structure parameters and hyperparameters of comparison deep reinforcement learning models are also determined by random search and grid search.

The comparison process includes the following aspects:

Firstly, the reward value and return value of each reinforcement learning algorithm during operation are compared under the same environment setting. Then, in terms of credibility, the significance test is used to calculate the significance of the return value accumulated by MATD3 relative to other methods, which can study whether MATD3 can be considered superior to other methods under extremely high credibility. Finally, the iterative optimization experiment is carried out on the trained and saved optimization model. And the machining parameters are iteratively optimized from the same starting point, and the optimization trajectory is observed to verify the superiority of MATD3 from the perspective of optimization objectives.

All comparison experiments are conducted using Pytorch with i5–12500H CPU, NVIDIA RTX 3060 and 16 G RAM. Furthermore, the machining parameters optimized by the proposed MATD3 are verified by actual milling machining to confirm the efficacy of the proposed method for real scenario applications. Detailed descriptions of the four groups of comparison models described above are shown in Appendix 2.

4.3. Results analysis on multi-task time series estimation models

4.3.1. Comparison between the proposed BMMoE-TE model and the single-task models

After the proposed BMMoE-TE model and the single-task RCNN-BiLSTM-TE model are trained to be stable, the comparison results of the evaluation metrics of each estimation task of the models are shown in Fig. 12. The CNC machining dataset comprises 11 tasks. Compared with the single-task model, the proposed BMMoE-TE model improves R^2 by 10 % −20 %, RMSE and MAE by about 10 % −40 % for each task. Based

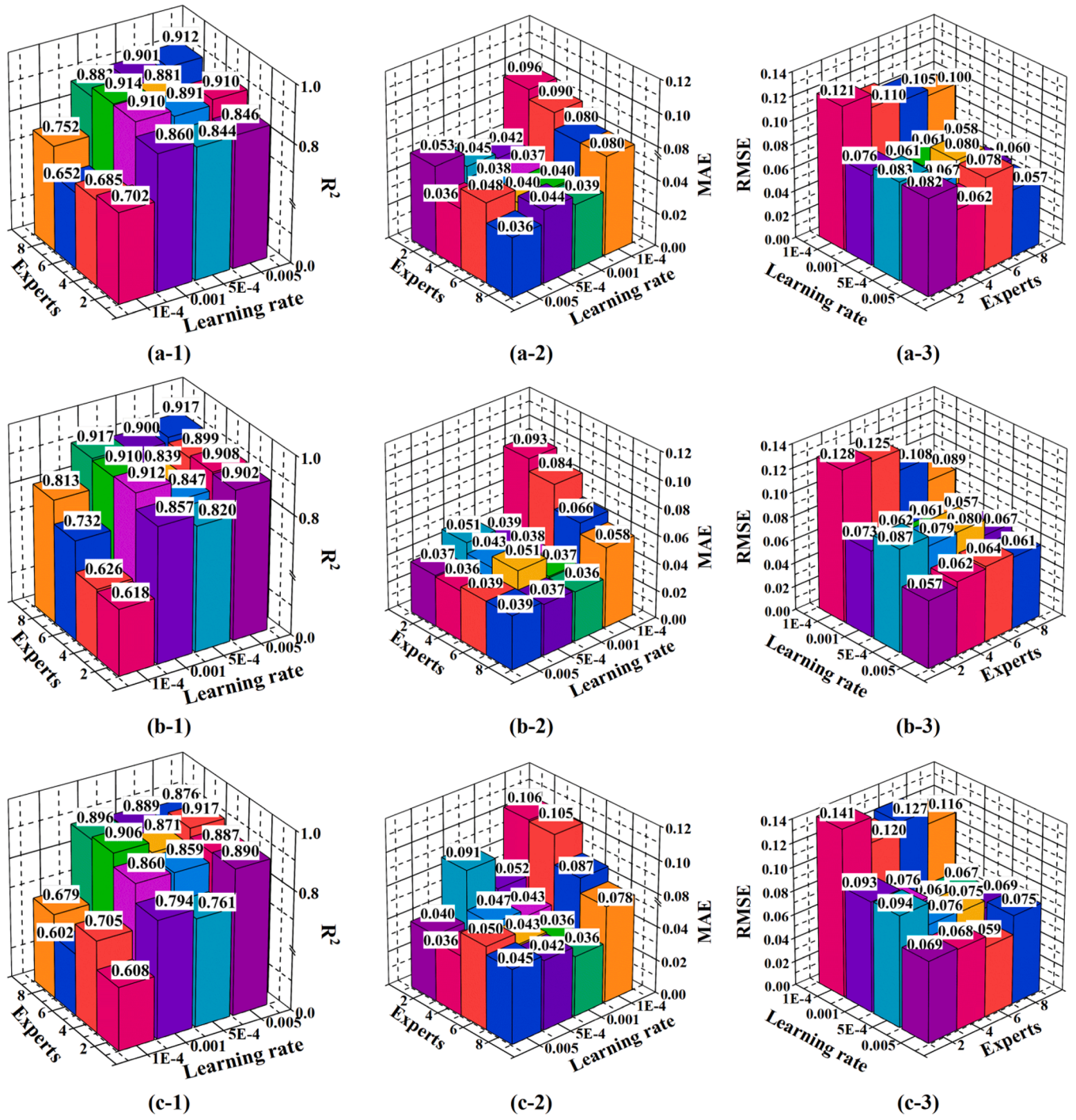

Fig. 9. Comparison of BMMoE-TE tuning parameters. (a) Batch size of 16; (b) Batch size of 32; (c) Batch size of 64.

Table 5
MATD3 model tuning parameter range.

Model structure	Parameters	Parameter space
Learning rate	Actor learning rate: Critic learning rate	{①[0.001:0.001], ②[0.001:0.005],③ [0.0005:0.001], ④[0.005:0.005]}
Discount factor	Gamma	{0.92,0.95,0.98}
Mini-batch size	Batch size	{16,32,64}

on the overall estimation results, the estimation performance of the proposed model is significantly better than that of the single-task estimation model, which means it is more suitable for supporting the virtual environment of DRL.

4.3.2. Comparison between BMMoE-TE (RCNN-BiLSTM) and the multi-task models

Fig. 13 displays the evaluation metrics curves generated by training each model for 600 epochs. The result of the quality indicators is the average of the evaluation metrics of the 10 quality indicators, and the result of the efficiency indicator corresponds to the evaluation metrics value of the completion time. Its results indicate that the proposed BMMoE-TE (RCNN-BiLSTM) model outperforms the comparison model in terms of accurately estimating future batch machining quality indicators and efficiency indicators.

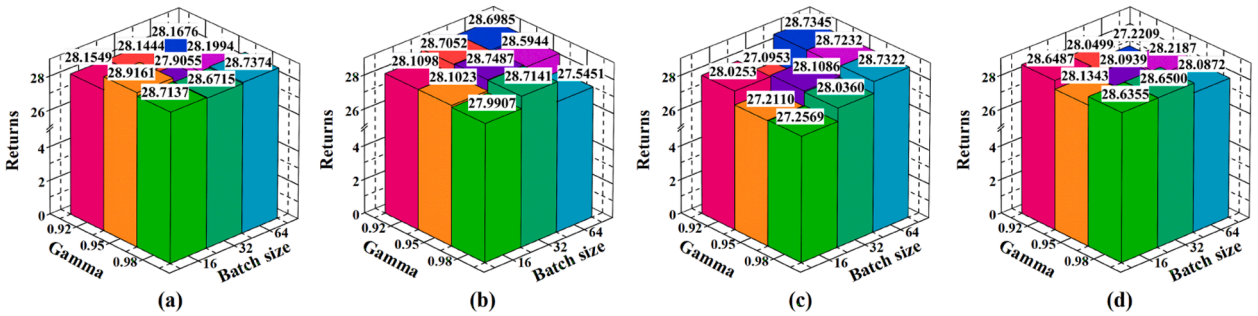

Fig. 10. Comparison of MATD3 tuning parameter returns. (a) learning rate group ①; (b) learning rate group ②; (c) learning rate group ③; (d) learning rate group ④.

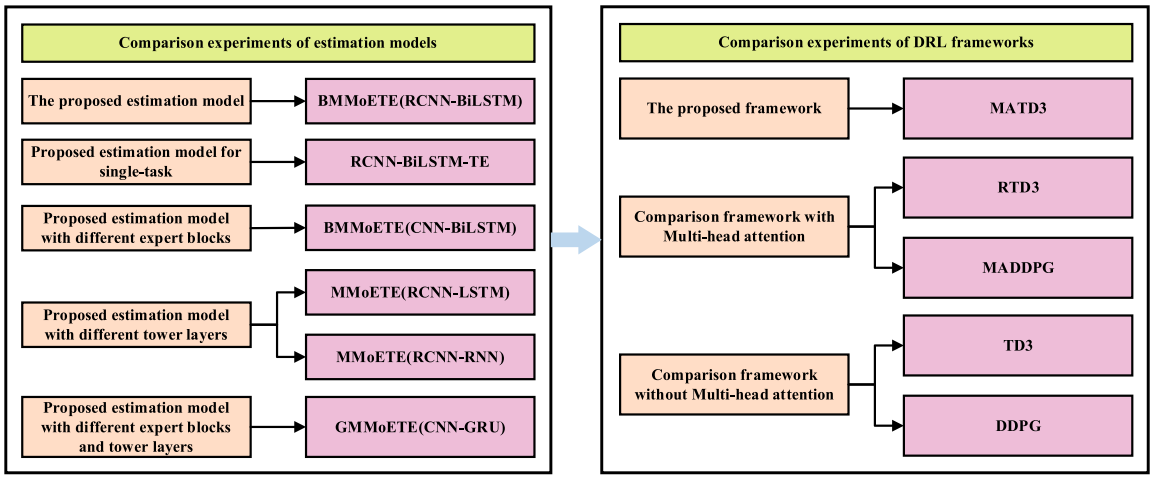

Fig. 11. Comparison experiments of different estimation models and DRL frameworks.

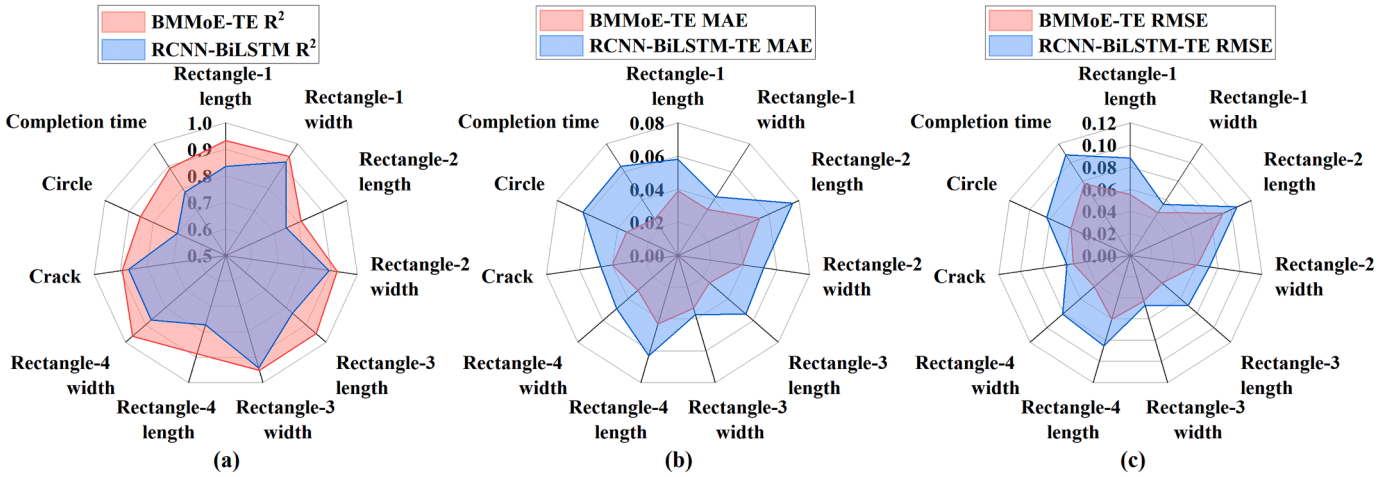

Fig. 12. Comparison between the proposed BMMoE-TE model and the single-task model. (a) Comparison of R^2; (b) Comparison of MAE; (C) Comparison of RMSE.

The trained models are tested. And the mean and standard deviation (Std) obtained from ten independent experiments with different random seeds were used as the final experimental results to quantify the improvement of the proposed BMMoE-TE (RCNN-BiLSTM) model relative to other models. The comparative results shown in Fig. 14 indicate that the performance of the BMMoE-TE (RCNN-BiLSTM) model improves relative to all other models after the model training is stabilized. Table 6 shows the results of the metrics improvement. Compared with other multi-task estimation models, the BMMoE-TE (RCNN-BiLSTM) model has an average increase of about 8 %−13 % on R^2. And the stability is significantly improved, MAE and RMSE increased by 8 % −33 % and 14 % −33 % on average respectively.

4.4. Results analysis on deep reinforcement learning

4.4.1. Comparison with other deep reinforcement learning methods

Four comparison deep reinforcement learning algorithms and the proposed MATD3, are used to perform 10 experiments and calculate the mean value. Each experiment trained 150 episodes, with each episode running up to 15 steps. The return in each episode and the single-step

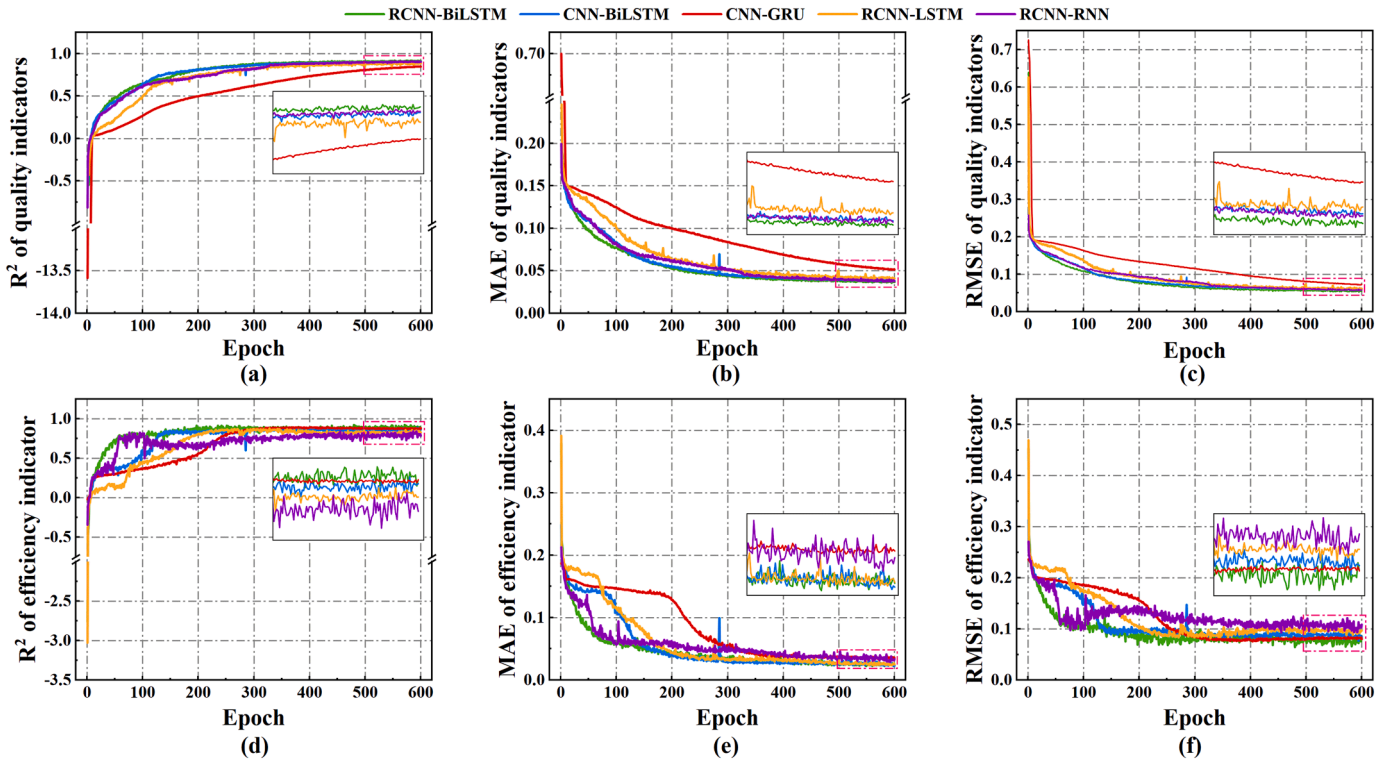

Fig. 13. Evaluation metric curves for 5 MMoE-TE models. (a)(b)(c) evaluation metrics of quality indicators; (d)(e)(f) evaluation metrics of efficiency indicators.

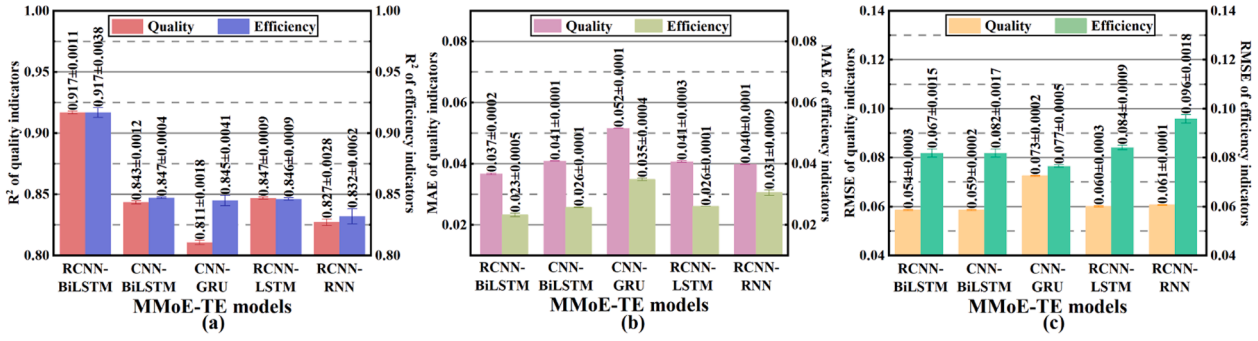

Fig. 14. Comparison results of evaluation metrics of 5 MMoE-TE models (a) Mean value of R^2; (b) Mean value of MAE; (c) Mean value of RMSE.

Table 6
Evaluation metrics improvement of BMMoE-TE (RCNN-BiLSTM) relative to comparative models.

MMoE-TE models	The type of estimation aims	Improvement rate of the R^2	Improvement rate of the MAE	Improvement rate of the RMSE
RCNN-BiLSTM	Quality	-	-	-
	Efficiency	-	-	-
CNN-BiLSTM	Quality	8.702 %	10.332 %	14.051 %
	Efficiency	8.222 %	9.643 %	17.543 %
CNN-GRU	Quality	13.099 %	28.937 %	27.632 %
	Efficiency	8.515 %	33.300 %	33.099 %
RCNN-LSTM	Quality	8.265 %	9.922 %	15.856 %
	Efficiency	8.354 %	22.725 %	18.433 %
RCNN-RNN	Quality	10.846 %	8.191 %	28.644 %
	Efficiency	10.187 %	23.974 %	27.768 %

running reward are shown in Fig. 15 (a) and (b), respectively. The reward is the value of the reward function value calculated according to Eq. (10) when the algorithm runs in one step, and the return is the cumulative result of the reward value in each episode. The figure displays the mean value obtained from the results of 10 experiments as a dark curve. The light-colored regions above and below the curve represent the maximum and minimum values recorded in each step or episode across the 10 experiments. To facilitate training, the elements in the replay buffer are processed to the same order of magnitude, so the reward value in the MDP is divided by 50. When the training is stable, the return per episode of proposed MATD3 is stable between 28.5–29.0, and the reward per episode is stable at about 1.92. However, the average

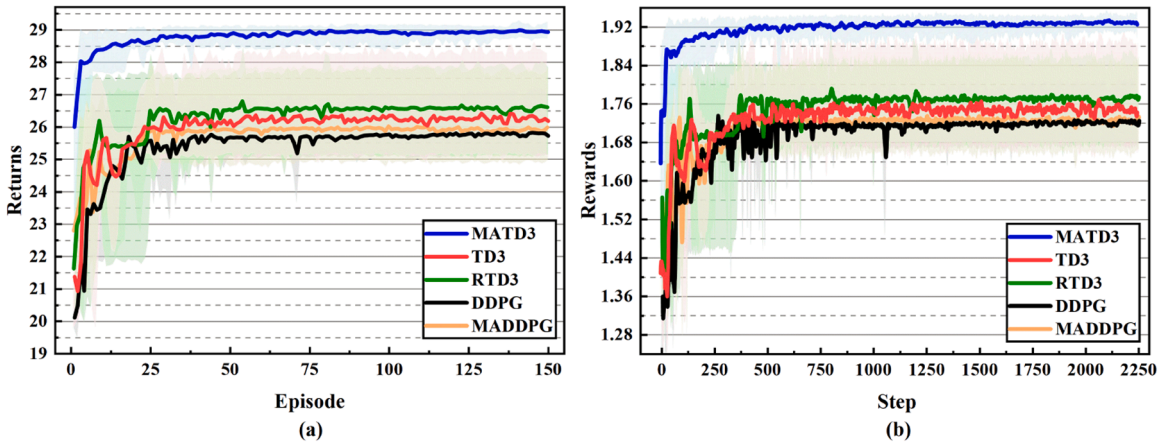

Fig. 15. The average rewards and returns of 5 algorithms. (a) average returns of 5 algorithms; (b) average rewards of 5 algorithms.

Table 7
Average return and average Std of last 50 episodes.

Algorithms	Average of the returns in the last 50 episodes	Std of the returns in last 50 episodes
MATD3	28.916	0.168
TD3	26.237	0.206
RTD3	26.541	0.203
DDPG	25.752	0.190
MADDPG	25.893	0.185

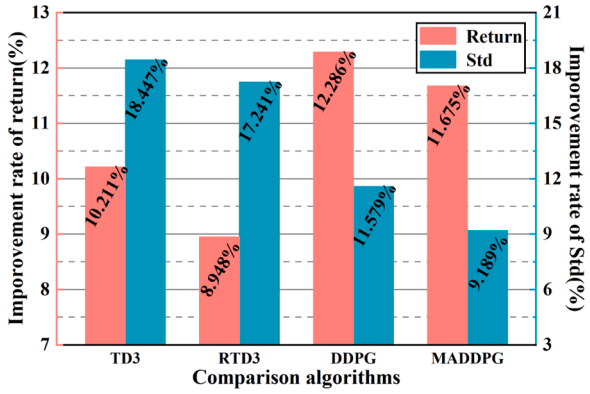

Fig. 16. The average improvement rate of return and Std of the last 50 episodes for four comparison models.

return per episode of other algorithms can only reach about 26, and the reward per episode is between 1.7 −1.8. The results indicate that the proposed MATD3 outperforms the other four algorithms in terms of stability, whether measured by single-step reward or return per episode.

To show differences in the performance and stability between the proposed MATD3 and other four algorithms in more details, the return and Std of each algorithm are compared. Firstly, taking the mean of each framework running 10 times as the object, the mean of the return of the last 50 episodes of the framework is calculated. Then, the std of the final 50 episodes is calculated based on the return of each framework running 10 times. The calculation results are shown in Table 7. The improvement rate of the average return and average std of the proposed MATD3 compared to other algorithms is shown in Fig. 16. According to the result, the std of the proposed MATD3 is improved by 9 % −18 %, and the average cumulative return of the proposed MATD3 is improved by 9 % −12 % compared to the other algorithms.

4.4.2. *Significance test*

The average statistical results of 10 times cannot completely represent the excellence of the proposed model statistically. Therefore, the significance test will be used to conduct a paired T-test on the return of the last 50 episodes for each experiment under different models. In the significance test, the null hypothesis means that the proposed method and the comparison method have equal optimization performance. Usually, it is necessary to set a significance level representing the probability threshold of rejecting the null hypothesis. In this paper, the significance level α is set to 0.05. If the P-value is less than 0.05, the null hypothesis is rejected, and the proposed MATD3 is considered to be significantly better than the comparison algorithm in this experiment. The significance test results of the proposed MATD3 relative to each comparison algorithm are shown in Fig. 17. The P-values calculated by the proposed MATD3 relative to each comparison algorithm under the results of 10 tests are all less than the significance level of 0.05. It shows that the proposed MATD3 statistically guarantees significant performance superiority. Therefore, the null hypothesis can be rejected and the proposed MATD3 can be considered to outperform other methods with 95 % (1–0.05) confidence.

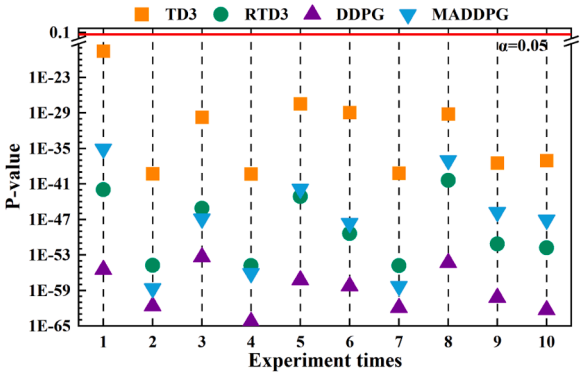

Fig. 17. Results of significance test.

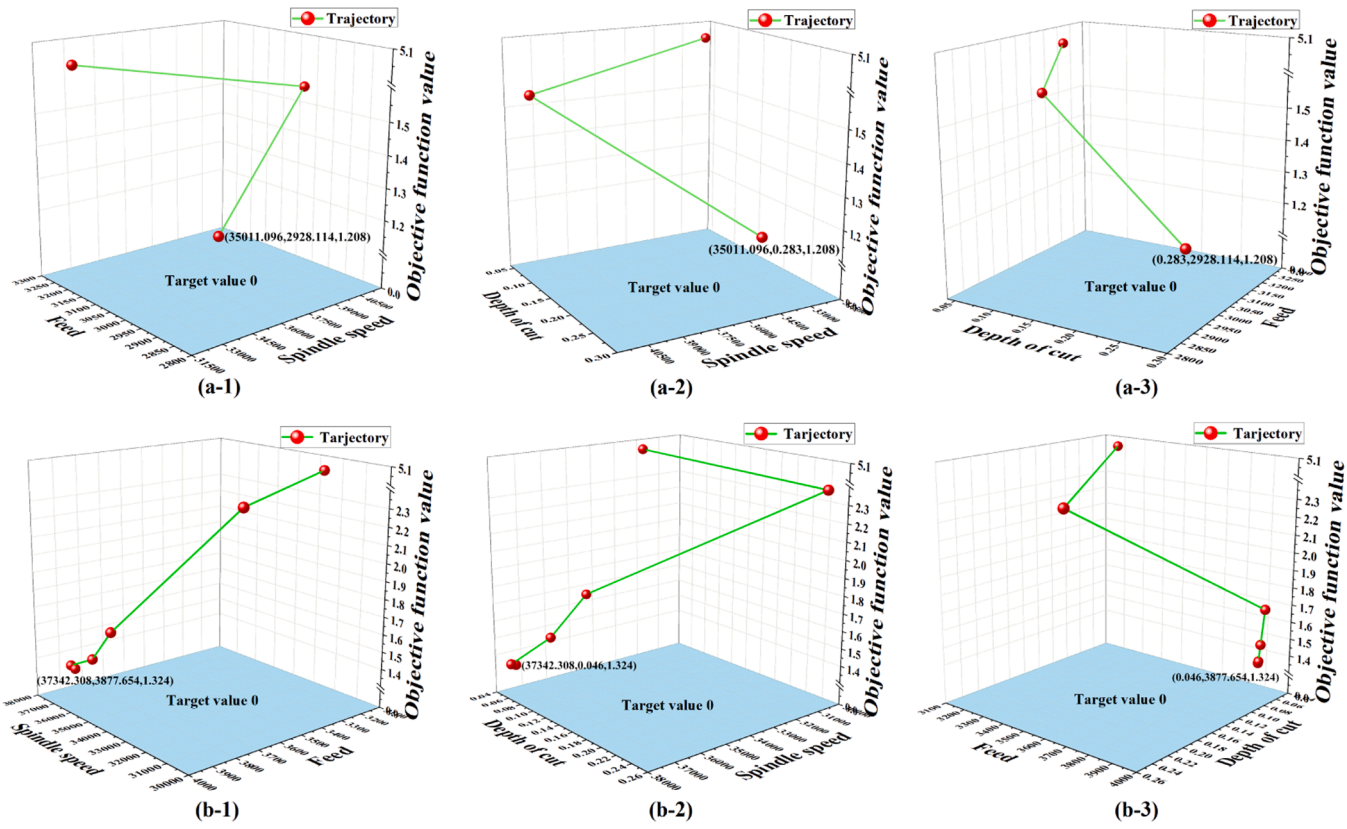

Fig. 18. Optimization trajectories for different algorithms. (a) Optimization trajectory of the proposed MATD3; (b) Optimization trajectory of TD3.

Table 8
The optimization results of the machining parameters.

Algorithms	Spindle speed (r/min)	Depth of cut (mm)	Feed (mm/min)	Iteration times	The optimized objective function value	Improvement rate of objective function value
MATD3	35011.096	0.283	2928.114	2	1.208	-
TD3	37342.308	0.046	3877.654	5	1.324	8.761 %
RTD3	37546.206	0.274	2854.882	6	1.332	9.309 %
DDPG	38911.911	0.043	3817.620	5	1.365	11.502 %
MADDPG	33517.275	0.286	2949.217	5	1.376	12.209 %

4.4.3. *Optimization trajectory comparison test*

To compare the machining parameter optimization policy learned by the proposed MATD3, the trained MATD3 and the other 4 comparison models are tested. The machining parameters are output during the iteration process and the three-dimension optimization trajectory has been drawn. The X-axis and Y-axis of the optimization trajectory are machining parameters. And the Z-axis is the objective function value corresponding to the optimized machining parameters obtained by each iteration in the optimization process, which is calculated by Eq. (3) from the machining indicators normalized between 0 and 1.

In the ideal case, each machining indicator reaches the ideal value, and the objective function value is 0. Therefore, each algorithm explores the optimization policy from the same starting point to the target value plane of $Z = 0$. The optimization trajectory of the proposed MATD3 and TD3 are shown in Fig. 18 (a) and (b), respectively. The optimization paths of the other 3 algorithms are shown in Appendix A3. In the optimization process, 3 machining parameters are considered at the same time. To facilitate the display, the algorithm search optimization policies form 3 sets of trajectory space by matching each two of the 3 machining parameters. The optimization results of the machining parameters of the 5 algorithms are shown in Table 8. It is noted that according to the algorithm principle, the machining parameters of the next workpiece outside the window can be recommended based on the time series window data. Because the closer it is to the window data, the more accurate the estimation result. Therefore, only the first set of machining parameters outside the window is recommended for workpiece machining to ensure the performance of the recommended parameters. The result indicates that the proposed MATD3 algorithm has reached the objective function value of 1.208 in only 2 iterations, which is closer to the target value plane of $Z = 0$ than the objective function values obtained by other 4 algorithms. The objective function value obtained in the proposed MATD3 is improved by 9 % −12 % compared with the other 4 algorithms, and the optimization effect of the proposed framework is improved. In addition, the other 4 algorithms may require a

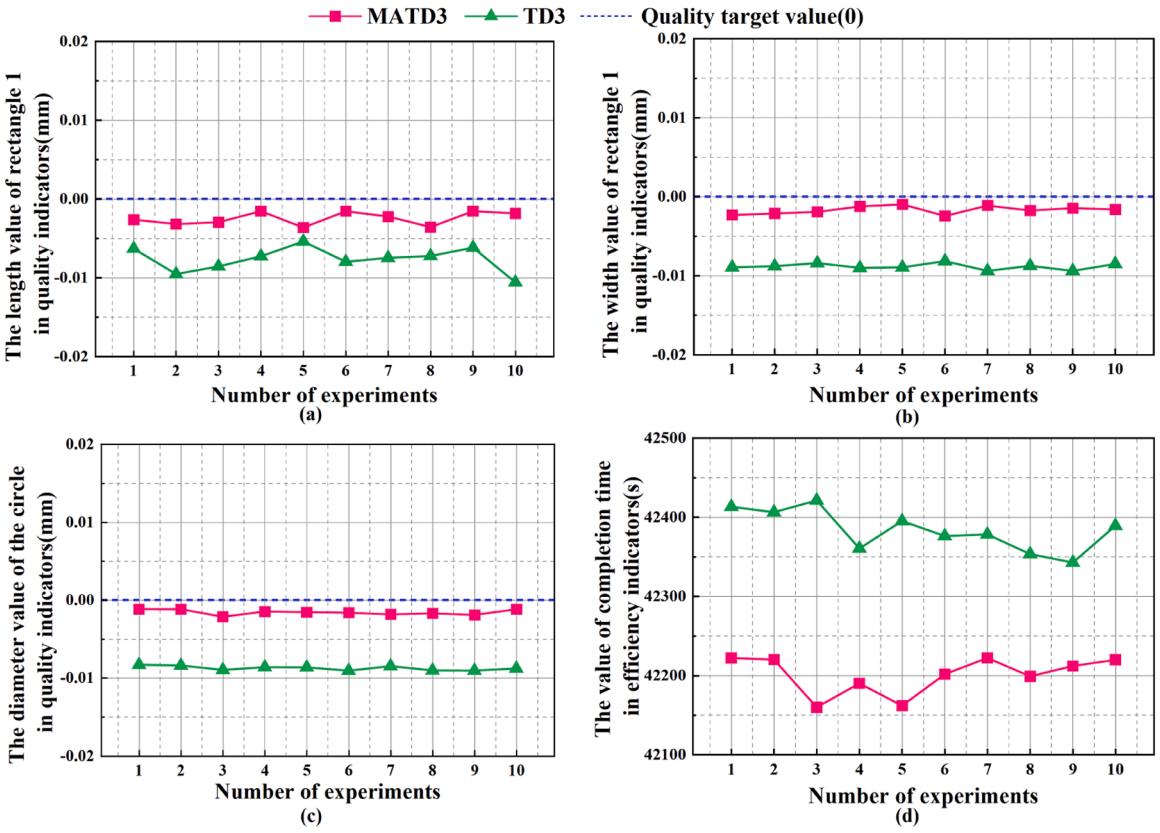

Fig. 19. Comparison results of machining indicators. (a) Comparison results in the length value of rectangle 1; (b) Comparison results in the width value of rectangle 1; (c) Comparison results of the circle; (d) Comparison results of completion time.

greater number of iterations to explore regions that are closer to the target value. Therefore, the proposed MATD3 has significant advantages in both optimization efficiency and performance.

4.5. Optimized parameter superiority verification

According to the results shown in Table 8, the optimized machining parameters of the workpiece of future batch are verified by actual milling machining. Due to the limitation of machining conditions, TD3 with excellent performance in the optimization policy exploration stage was selected as the comparison algorithm in the verification experiment. In order to verify the quality of the recommended machining parameters, the machining parameters recommended by MATD3 and TD3 were set on a CNC machine to produce 10 workpieces respectively, and the results of their machining indicators and target values were analyzed. After each workpiece is processed, the machining features of the machined workpieces are measured, and the dimensional error values are calculated as the machining quality indicator values. And the completion time is recorded as the machining efficiency indicator values.

The maximum dimensional error value of each machining workpiece of the thin-walled parts machined in this paper is $\pm$ 0.015 mm. The machining indicator values obtained from 10 experiments are shown in Fig. 19. The dimensional error values of the circle, the length of rectangle 1, and the width of rectangle 1 in the machining quality indicators and the completion time corresponding to the machining efficiency indicators are taken as examples. The comparison of the regression results of the remaining machining indicators are shown in Appendix A4. For the machining quality indicators, the closer they are to 0, the smaller the dimensional error value and the better the optimized machining parameters. The results show that the machining quality indicator values corresponding to the machining parameters optimized by the MATD3 is closer to 0. Additionally, the machining efficiency indicator values for MATD3 is lower than that of the TD3 and RTD3. In summary, MATD3 offers advantages in collaborative improvement of quality and efficiency, as well as the optimization of machining parameters.

5. Conclusions

In this paper, the MTE-BDRL framework is designed to machining parameter optimization with correlation among multiple objectives. The approach combines the multi-task parallel joint estimation with improved reinforcement learning MATD3 to improve the accuracy of optimized machining parameters for the collaborative optimization of quality and efficiency multivariate machining indicators. Here, some conclusions are summarized below:

1) The multi-task parallel joint estimation model based on BMMoE-TE with RCNN as the expert sharing layer and BiLSTM as the tower layer is designed to jointly estimate machining indicators of the future workpieces.

2) The reinforcement learning virtual environment with the joint estimation model is constructed to improve the accuracy of optimized machining parameters for the collaborative optimization of quality and efficiency multivariate machining indicators. By adding multi-head attention and dynamic weight adjustment mechanisms, the stability of algorithm training is improved.

3) The proposed framework is validated through various numerical studies using a real case study of antenna thin-walled component data. The experimental results show that the proposed BMMoE-TE provides better estimation than single task models as well as other multi-task estimation models.

4) Compared with the traditional deep reinforcement learning algorithm, the ability of MATD3 to accumulate returns increased by 9 % −12 % in the same optimization aim and virtual environment. The return value is calculated by the reward function constructed under the guidance of the optimization objective function, which is the direct expression of the optimization objective in the agent optimization process. Therefore, it can be seen that the optimization effect of MATD3 is improved by 9 % −12 %. In addition, the standard deviation of MATD3 return value is reduced by about 9 % −18 % under multiple experimental results. Therefore, MATD3 has better optimization effect and higher stability.

Although the framework proposed in this paper has achieved relatively good machining parameter optimization accuracy, it can still be further improved. For future work, multi-agent reinforcement learning and evolutionary learning can be introduced to enhance the ability of the framework to avoid local optimal solutions. Simultaneously, further exploration can be made in the reward function aspect to enhance the decoupling ability of the algorithm in solving multi-objective optimization problems. So that the optimized machining parameters can be better to coupling improve the machining quality and efficiency of the workpieces.

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CRediT authorship contribution statement

Pei Wang: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Sheng Yang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Xun Xu:** Writing – review & editing, Writing – original draft, Conceptualization. **Haizhen Tao:** Writing – original draft, Validation, Investigation, Data curation. **Yixin Cui:** Writing – original draft, Validation, Investigation, Data curation.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pei Wang reports financial support was provided by National Natural Science Foundation of China. Pei Wang reports financial support was provided by Natural Science Basic Research Plan in Shaanxi Province of China.

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Appendix 1 Pseudocode of the algorithms

A1–1 Pseudocode of the BMMoE-TE.

Input: time window features data X_e and indicators data Y_e. Batch size, epoch, learning rate.

Output: Multi-task time series estimation results.

I Initialization: Multi-task time series estimation parameters $\Theta = \{\theta^1, \ldots, \theta^M\}$,where $\theta^m = \{\theta_s, W_g^m, \theta_B^m\}$.

II Training model:

1. for 1 to EPOCH
2. Calculate the result of RCNN $X_{e'(t)} = R(X_{e(t)}, \theta_s)$.
3. Calculate the gated network $g^m(X_{e(t)}) = softmax(W_g^m X_{e(t)})$.
4. Calculate the underlying shared network $F^m(X_{e(t)}) = \sum_{i=1}^{n} g^m(X_{e(t)})_i (X_{e'(t)})_i$.
5. Output the tower layer time series estimation result $\widetilde{\omega_{t+1}^m}$.
6. Calculate the loss of the dynamic weighted balance $L = \sum_m^M \lambda_m L_m$.
7. Update the parameters of each network $\theta^m \leftarrow \theta^m - \vartheta^R \left(\frac{\partial \lambda_m L_m}{\partial \theta^m}\right)$.
8. End for

A1–2 Pseudocode of the MATD3.

Input: The capacity of the replay buffer $\mathscr{B}$, mini-batch size $\mathcal{N}$, discount factor γ, the learning rate of actor network $\vartheta^{\boldsymbol{A}}$, the learning rate of critic network ϑ^{c}, the weight of soft updating φ.

I **Initialization:** The actor evaluation network, the critic evaluation network and parameters of them $\theta_E^{\boldsymbol{A}}$, $\theta_E^{c(1)}$, $\theta_E^{c(2)}$; The actor target network, the critic target network and parameters of them $\theta_T^{\boldsymbol{A}}$, $\theta_T^{c(1)}$, $\theta_T^{c(2)}$.

II **Training model:**

1. for episode=1 to EPISODE do
2. randomly initialize the initial sequential state $\boldsymbol{S}_1$
3. for step=1 to STEP do
4. Input state $\boldsymbol{S}_\tau$ into the actor evaluation network, and select sequential action $\boldsymbol{A}_\tau = \delta\left(MA\left(\boldsymbol{S}_\tau \overrightarrow{\otimes \,\&\, \| \, \boldsymbol{S}_\tau^{MA}}\right)\middle|\theta_E^{\boldsymbol{A}}\right) + \xi_\tau$ with the multi-attention mechanism
5. Execute sequential action $\boldsymbol{A}_\tau$ in virtual environment, getting the reward r_τ and the next sequential state $\boldsymbol{S}_{\tau+1}$
6. Store experiences in the replay buffer $(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau, r_\tau, \boldsymbol{S}_{\tau+1})$
7. Sample a random mini-batch of $\mathcal{N}$ experiences $(\boldsymbol{S}_\tau, \boldsymbol{A}_\tau, r_\tau, \boldsymbol{S}_{\tau+1})$ from the replay buffer
8. Calculate the temporal-difference target β_τ
9. The loss $L(\theta_E^{c(1)}, \theta_E^{c(2)})$ is minimized by Eq. (13) to update the critic evaluation network
10. The strategy gradient $\nabla_{\theta_E^{\boldsymbol{A}}} J(\theta_E^{\boldsymbol{A}})$ is boosted by Eq. (14) to update the actor evaluation network
11. Update the target network: $\begin{cases} \theta_T^{c(1)} \leftarrow \varphi\theta_E^{c(1)} + (1-\varphi)\theta_T^{c(1)} \\ \theta_T^{c(2)} \leftarrow \varphi\theta_E^{c(2)} + (1-\varphi)\theta_T^{c(2)} \\ \theta_T^{\boldsymbol{A}} \leftarrow \varphi\theta_E^{\boldsymbol{A}} + (1-\varphi)\theta_T^{\boldsymbol{A}} \end{cases}$
12. end for
13. end for

Appendix 2 Descriptions of the comparison models

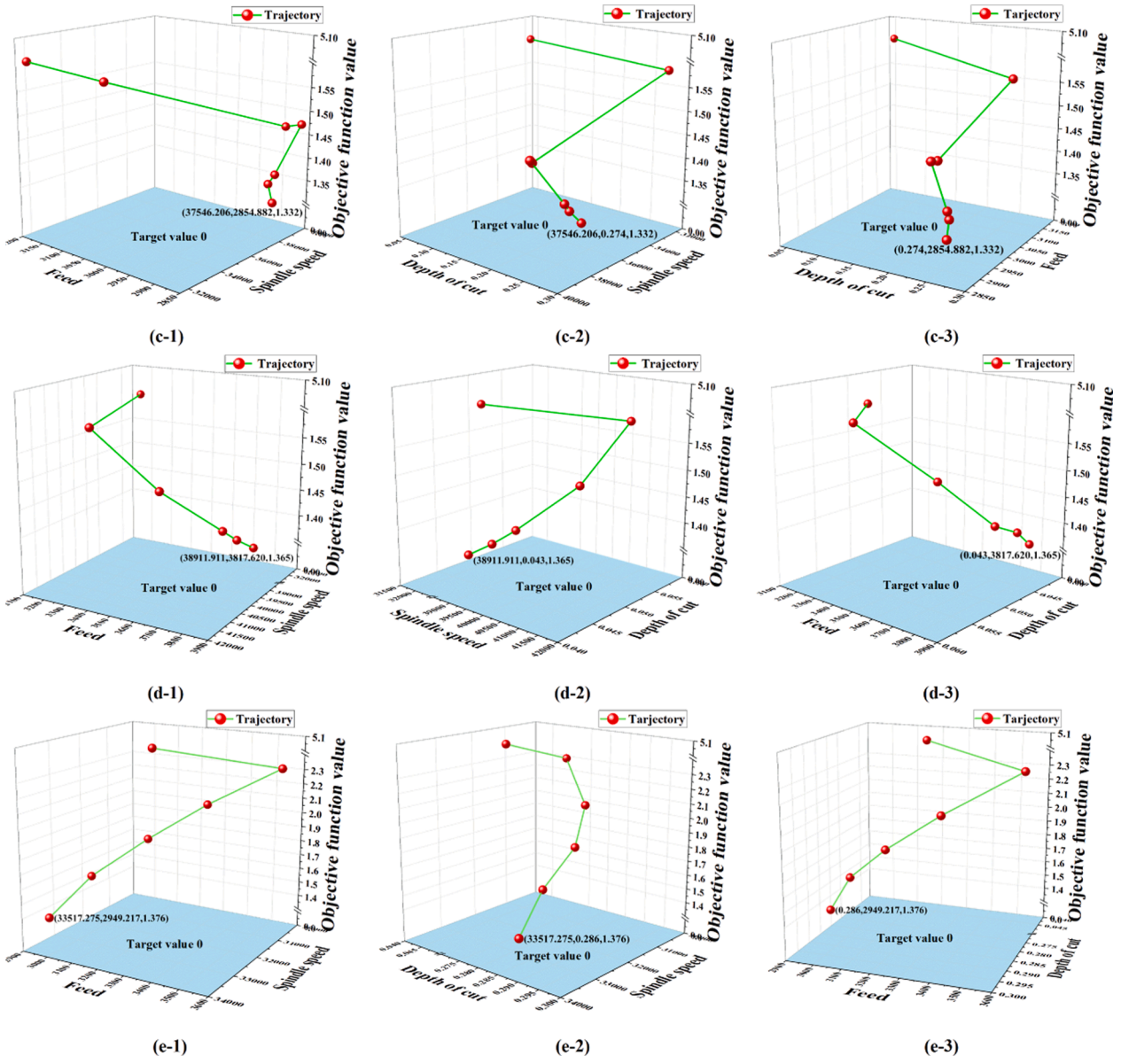
A2–1 Descriptions of the comparison multi-task time series models.

Comparison framework	Comparison model	**Description**
Comparison framework for single-task estimation	RCNN-BiLSTM-TE	Single-task learning with the same RCNN as the expert sharing layer of BMMoE-TE(RCNN-BiLSTM), and the BiLSTM is set to 8 −32 −1.
Comparison framework with different expert sharing layer	BMMoE-TE (CNN-BiLSTM)	The tower layer is the same as in BMMoE-TE(RCNN-BiLSTM), but without residuals in the expert sharing layer.
	MMoE-TE (RCNN-LSTM)	The expert sharing layer is the same as in BMMoE-TE(RCNN-BiLSTM), and the tower layer of LSTM is set to 8–16–1.
Comparison framework with different tower layer	MMoE-TE (RCNN-RNN)	The expert sharing layer is the same as in BMMoE-TE(RCNN-BiLSTM), and the tower layer of RNN is set to 8–16–1.
Comparison framework with different expert sharing layer and tower layer	GMMoE-TE (CNN-GRU)	The tower layer of GRU is set to 8 −16 −1, and without residuals in the expert sharing layer.

A2–2 Descriptions of the deep reinforcement learning models.

Comparison framework	Comparison model	**Description**
Comparison framework without multi-head attention	TD3	The critic networks are the same as the MATD3, but without multi-head attention in the actor network.
	DDPG	There is only one critic network, and without multi-head attention in the actor network.
Comparison framework with multi-head attention	RTD3	The critic networks are the same as the MATD3, and with a linear layer branch to each multi-head attention module in the actor network.
	MADDPG	There is only one critic network, but the actor network is the same as the MATD3

Appendix 3 Algorithm optimization trajectories



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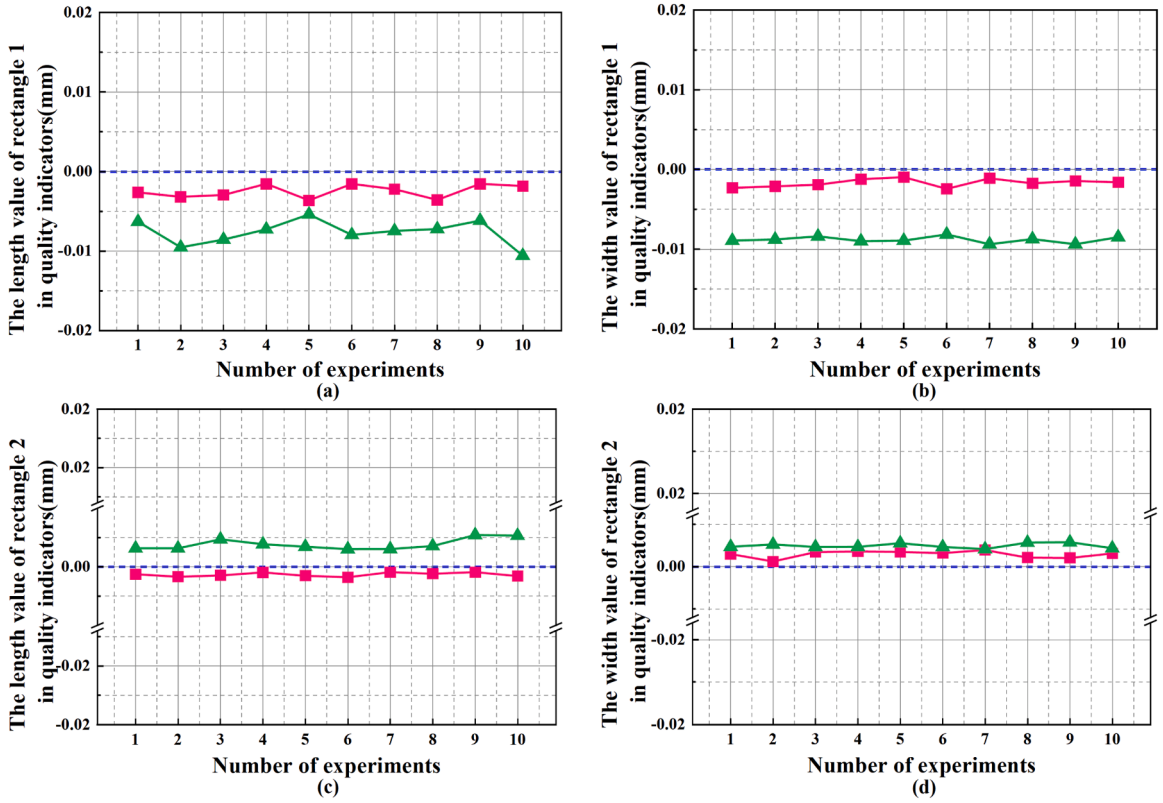
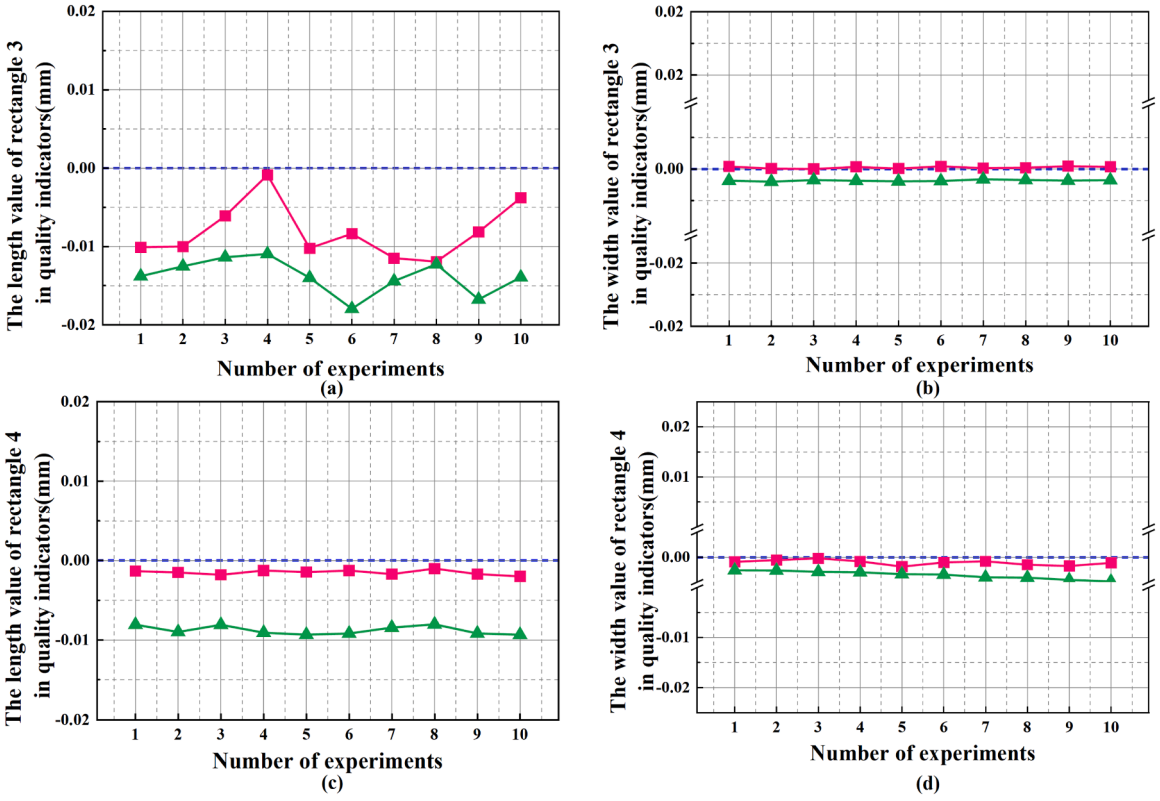
A3. Algorithm optimization trajectories. (c) Optimization trajectory of RTD3; (d) Optimization trajectory of DDPG; (e) Optimization trajectory of MADDPG.

Appendix 4 Comparison results of machining indicators



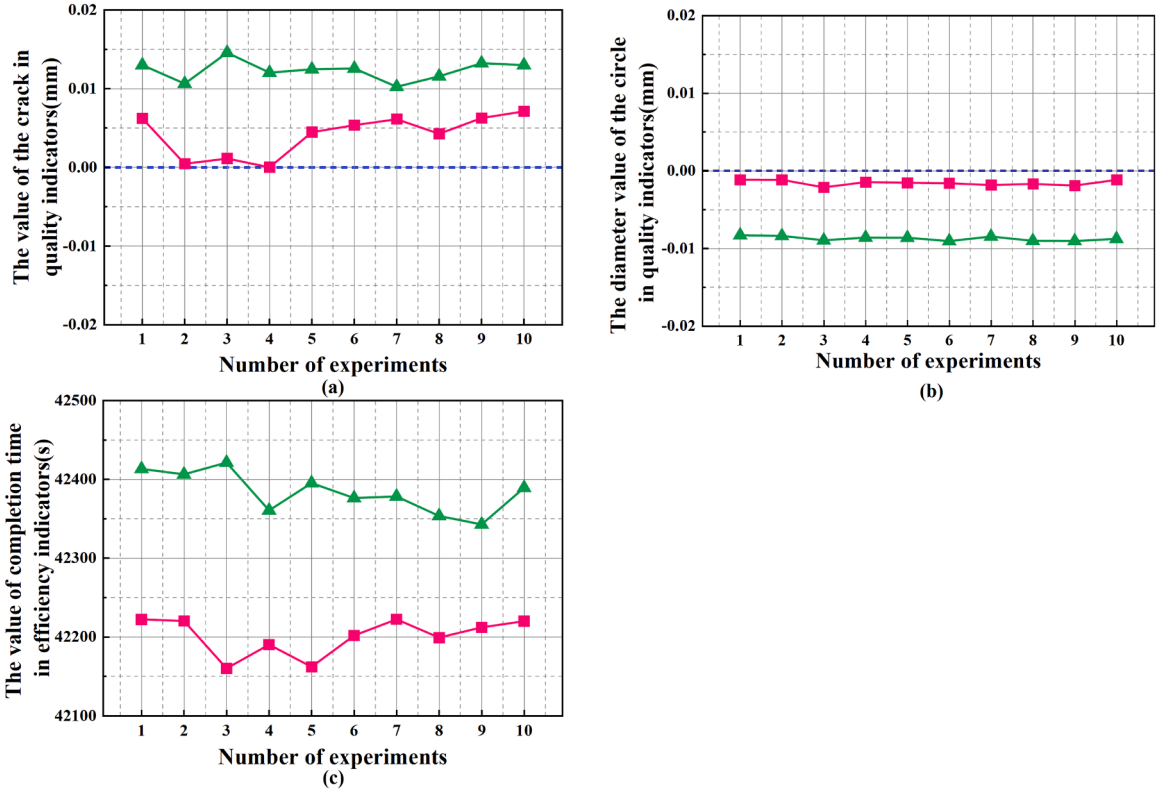
.

A4–1. Comparison results of the rectangle 1 and rectangle 2. (a) Comparison results in the length value of rectangle 1; (b) Comparison results in the width value of rectangle 1; (c) Comparison results in the length value of rectangle 2; (d) Comparison results in the width value of rectangle 2.



.

A4–2. Comparison results of the rectangle 3 and rectangle 4. (a) Comparison results in the length value of rectangle 3; (b) Comparison results in the width value of rectangle 3; (c) Comparison results in the length value of rectangle 4; (d) Comparison results in the width value of rectangle 4.



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A4–3. Comparison results of the crack, circle and completion time. (a) Comparison results in the value of the crack; (b) Comparison results in the value of the circle; (c) Comparison results in the value of completion time.

References

[1] Zhu Z, Xi X, Xu X, Cai Y. Digital Twin-driven machining process for thin-walled part manufacturing. J Manuf Syst 2021;59:453–66. https://doi.org/10.1016/j.jmsy.2021.03.015.

[2] Wang Z, Lu J, Chen C, Ma J, Liao X. Investigating the multi-objective optimization of quality and efficiency using deep reinforcement learning. Appl Intell 2022;52:12873–87. https://doi.org/10.1007/s10489-022-03326-5.

[3] Li W, He S, Mao X, Li B, Qiu C, Yu J, et al. Multi-agent evolution reinforcement learning method for machining parameters optimization based on bootstrap aggregating graph attention network simulated environment. J Manuf Syst 2023;67:424–38. https://doi.org/10.1016/j.jmsy.2023.02.015.

[4] Lu F, Zhou G, Zhang C, Liu Y, Chang F, Xiao Z. Energy-efficient multi-pass cutting parameters optimisation for aviation parts in flank milling with deep reinforcement learning. Robot Comput-Integr Manuf 2023;81:102488. https://doi.org/10.1016/j.rcim.2022.102488.

[5] Tercan H, Meisen T. Machine learning and deep learning based predictive quality in manufacturing: a systematic review. J Intell Manuf 2022;33:1879–905. https://doi.org/10.1007/s10845-022-01963-8.

[6] Lee J, Noh S, Kim H-J, Kang Y-S. Implementation of cyber-physical production systems for quality prediction and operation control in metal casting. Sensors 2018;18:1428. https://doi.org/10.3390/s18051428.

[7] Wang P, Bu F, Kong X, Chang J, Cui Y, Zhang A. Quality–efficiency coupling prediction and monitoring-based process optimization of thin plate parts with multi-machining feature. Int J Comput Integr Manuf 2023:1–31. https://doi.org/10.1080/0951192X.2023.2264831.

[8] Li B-H, Zhao L-P, Yao Y-Y. Multiconditional machining process quality prediction using deep transfer learning network. Adv Manuf 2023;11:329–41. https://doi.org/10.1007/s40436-022-00415-z.

[9] Sun Y-N, Qin W, Xu H-W, Tan R-Z, Zhang Z-L, Shi W-T. A multiphase information fusion strategy for data-driven quality prediction of industrial batch processes. Inf Sci 2022;608:81–95. https://doi.org/10.1016/j.ins.2022.06.057.

[10] Chen H, Yang Y, Shao C. Multi-task learning for data-efficient spatiotemporal modeling of tool surface progression in ultrasonic metal welding. J Manuf Syst 2021;58:306–15. https://doi.org/10.1016/j.jmsy.2020.12.009.

[11] Liu C, Ni J, Wan P. An accurate prediction method of multiple deterioration forms of tool based on multitask learning with low rank tensor constraint. J Manuf Syst 2021;58:193–204. https://doi.org/10.1016/j.jmsy.2020.11.018.

[12] Wang P, Qu H, Zhang Q, Xu X, Yang S. Production quality prediction of multistage manufacturing systems using multi-task joint deep learning. J Manuf Syst 2023;70:48–68. https://doi.org/10.1016/j.jmsy.2023.07.002.

[13] Wang Y, Qin B, Liu K, Shen M, Niu M, Han L. A new multitask learning method for tool wear condition and part surface quality prediction. IEEE Trans Ind Inf 2021;17:6023–33. https://doi.org/10.1109/TII.2020.3040285.

[14] Chen X, Li C, Tang Y, Li L, Li H. Energy efficient cutting parameter optimization. Front Mech Eng 2021;16:221–48. https://doi.org/10.1007/s11465-020-0627-x.

[15] Zhang X, Pan T, Ma A, Zhao W. High efficiency orientated milling parameter optimization with tool wear monitoring in roughing operation. Mech Syst Signal Process 2022;165:108394. https://doi.org/10.1016/j.ymssp.2021.108394.

[16] Wang W, Tian G, Chen M, Tao F, Zhang C, AI-Ahmari A, et al. Dual-objective program and improved artificial bee colony for the optimization of energy-conscious milling parameters subject to multiple constraints. J Clean Prod 2020;245:118714. https://doi.org/10.1016/j.jclepro.2019.118714.

[17] Wang Q, Li P, Rocca P, Li R, Tan G, Hu N, et al. Interval-based tolerance analysis method for petal reflector antenna with random surface and deployment errors. IEEE Trans Antennas Propagat 2023;71:8556–69. https://doi.org/10.1109/TAP.2023.3314097.

[18] Xiao Y, Jiang Z, Gu Q, Yan W, Wang R. A novel approach to CNC machining center processing parameters optimization considering energy-saving and low-cost. J Manuf Syst 2021;59:535–48. https://doi.org/10.1016/j.jmsy.2021.03.023.

[19] Li B, Tian X, Zhang M. Modeling and multi-objective optimization method of machine tool energy consumption considering tool wear. Int J Precis Eng Manuf-Green Tech 2022;9:127–41. https://doi.org/10.1007/s40684-021-00320-z.

[20] Jiang Y, Chen J, Zhou H, Yang J, Hu P, Wang J. Contour error modeling and compensation of CNC machining based on deep learning and reinforcement learning. Int J Adv Manuf Technol 2022;118:551–70. https://doi.org/10.1007/s00170-021-07895-6.

[21] Liu C, Li Y, Huang C, Zhao Y, Zhao Z. A meta-reinforcement learning method by incorporating simulation and real data for machining deformation control of finishing process. Int J Prod Res 2023;61:1114–28. https://doi.org/10.1080/00207543.2022.2027041.

[22] Lu J, Chen Z, Liao X, Chen C, Ouyang H, Li S. Multi-objective optimization for improving machining benefit based on WOA-BBPN and a Deep Double Q-Network. Appl Soft Comput 2023;142:110330. https://doi.org/10.1016/j.asoc.2023.110330.

[23] Wang Y, Peng T, Wang W, Luo M. High-efficient view planning for surface inspection based on parallel deep reinforcement learning. Adv Eng Inform 2023;55:101849. https://doi.org/10.1016/j.aei.2022.101849.

[24] Huang J, Su J, Chang Q. Graph neural network and multi-agent reinforcement learning for machine-process-system integrated control to optimize production yield. J Manuf Syst 2022;64:81–93. https://doi.org/10.1016/j.jmsy.2022.05.018.

[25] Xiao Q, Li C, Tang Y, Li L. Meta-reinforcement learning of machining parameters for energy-efficient process control of flexible turning operations. IEEE Trans Autom Sci Eng 2021;18:5–18. https://doi.org/10.1109/TASE.2019.2924444.

[26] Zimmerling C, Poppe C, Stein O, Kärger L. Optimisation of manufacturing process parameters for variable component geometries using reinforcement learning. Mater Des 2022;214:110423. https://doi.org/10.1016/j.matdes.2022.110423.

[27] Li W, Li B, He S, Mao X, Qiu C, Qiu Y, et al. A novel milling parameter optimization method based on improved deep reinforcement learning considering machining cost. J Manuf Process 2022;84:1362–75. https://doi.org/10.1016/j.jmapro.2022.11.015.

[28] Han Y, Peng H, Mei C, Cao L, Deng C, Wang H, et al. Multi-strategy multi-objective differential evolutionary algorithm with reinforcement learning. Knowl-Based Syst 2023;277:110801. https://doi.org/10.1016/j.knosys.2023.110801.

[29] Li K, Zhang T, Wang R. Deep reinforcement learning for multiobjective optimization. IEEE Trans Cyber 2021;51:3103–14. https://doi.org/10.1109/TCYB.2020.2977661.

[30] Marukatat S. Tutorial on PCA and approximate PCA and approximate kernel PCA. Artif Intell Rev 2023;56:5445–77. https://doi.org/10.1007/s10462-022-10297-z.

[31] Hilali Y, Bourihane O. A meshfree Hermite point interpolation method for buckling and post-buckling analysis of thin plates. Eng Comput 2022;38:3171–90. https://doi.org/10.1007/s00366-021-01457-w.

[32] Zhang Y, Xin Y, Liu Z, Chi M, Ma G. Health status assessment and remaining useful life prediction of aero-engine based on BiGRU and MMoE. Reliab Eng Syst Saf 2022;220:108263. https://doi.org/10.1016/j.ress.2021.108263.

[33] Yang Z, Xi Z, Zhang T, Guo W, Zhang Z, Li H-C. CMR-CNN: Cross-Mixing Residual Network for Hyperspectral Image Classification. IEEE J Sel Top Appl Earth Obs Remote Sens 2022;15:8974–89. https://doi.org/10.1109/JSTARS.2022.3213865.

[34] Chien S-H, Sencer B, Ward R. Accurate prediction of machining cycle times and feedrates with deep neural networks using BiLSTM. J Manuf Syst 2023;68:680–6. https://doi.org/10.1016/j.jmsy.2023.05.020.

[35] Cheng H, Kong X, Chen G, Wang Q, Wang R. Transferable convolutional neural network based remaining useful life prediction of bearing under multiple failure behaviors. Measurement 2021;168:108286. https://doi.org/10.1016/j.measurement.2020.108286.

[36] Wang X, Zhang L, Liu Y, Zhao C, Wang K. Solving task scheduling problems in cloud manufacturing via attention mechanism and deep reinforcement learning. J Manuf Syst 2022;65:452–68. https://doi.org/10.1016/j.jmsy.2022.08.013.

[37] Ke H, Wang J, Deng L, Ge Y, Wang H. Deep reinforcement learning-based adaptive computation offloading for MEC in heterogeneous vehicular networks. IEEE Trans Veh Technol 2020;69:7916–29. https://doi.org/10.1109/TVT.2020.2993849.

[38] Tufenkci S, Baykant Alagoz B, Kavuran G, Yeroglu C, Herencsar N, Mahata S. A theoretical demonstration for reinforcement learning of PI control dynamics for optimal speed control of DC motors by using Twin Delay Deep Deterministic Policy Gradient Algorithm. Expert Syst Appl 2023;213:119192. https://doi.org/10.1016/j.eswa.2022.119192.

[39] Liu S., Johns E., Davison A.J. End-To-End Multi-Task Learning With Attention. 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, CA, USA: IEEE; 2019, p. 1871–80. https://doi.org/10.1109/CVPR.2019.00197.

[40] Bergstra J, Bengio Y. Random search for hyper-parameter optimization. J Mach Learn Res 2012;13:281–305.

[41] Wang P, Qi J, Xu X, Yang S. Machining quality prediction of multi-feature parts using integrated multi-source domain dynamic adaptive transfer learning. Robot Comput-Integr Manuf 2024;90:102815. https://doi.org/10.1016/j.rcim.2024.102815.

[42] García-Pérez A, Ziegenbein A, Schmidt E, Shamsafar F, Fernández-Valdivielso A, Llorente-Rodríguez R, et al. CNN-based in situ tool wear detection: a study on model training and data augmentation in turning inserts. J Manuf Syst 2023;68:85–98. https://doi.org/10.1016/j.jmsy.2023.03.005.

[43] Liu Z, Zhang D, Jia W, Lin X, Liu H. An adversarial bidirectional serial–parallel LSTM-based QTD framework for product quality prediction. J Intell Manuf 2020;31:1511–29. https://doi.org/10.1007/s10845-019-01530-8.

[44] Shi Y, Zhao W, Li S, Li B, Sun X. Direct derivation scheme of DT-RNN algorithm for discrete time-variant matrix pseudo-inversion with application to robotic

manipulator. Appl Soft Comput 2023;133:109861. https://doi.org/10.1016/j.asoc.2022.109861.
[45] Liu X, Zhang B, Li X, Liu S, Yue C, Liang SY. An approach for tool wear prediction using customized DenseNet and GRU integrated model based on multi-sensor feature fusion. J Intell Manuf 2023;34:885–902. https://doi.org/10.1007/s10845-022-01954-9.
[46] Wang Z, Jin X, Zhang T, Li J, Yu D, Cheong KH, et al. Expert system-based multiagent deep deterministic policy gradient for swarm robot decision making. IEEE Trans Cyber 2022:1–11. https://doi.org/10.1109/TCYB.2022.3228578.
[47] Leng J, Lin Z, Zhou M, Liu Q, Zheng P, Liu Z, et al. Multi-layer parallel transformer model for detecting product quality issues and locating anomalies based on multiple time-series process data in Industry 4.0. J Manuf Syst 2023;70:501–13. https://doi.org/10.1016/j.jmsy.2023.08.013.